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Test – 22

Science & Technology

Topics covered:

- Space Technology
- Defence
- Electronics and Green Tech
- Information Technology (IT)+Robotics(A.I)
- Nuclear + Particle Science(Nano Tech)
- Current Affairs September 2021



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DEDICATED PRELIM PROGRAM(DPP)

UPSC-CSE PRELIMINARY TEST SERIES – 2021/2022

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TEST BOOKLET SERIES

Test Code:

DPP

**TEST BOOKLET
GENERAL STUDIES
TEST - 22**

E 22

Time Allowed: Two Hours

Maximum Marks : 200

INSTRUCTIONS

1. IMMEDIATELY AFTER THE COMMENCEMENT OF THE EXAMINATION, YOU SHOULD CHECK THAT THIS TEST BOOKLET DOES NOT HAVE ANY UNPRINTED OR TORN OR MISSING PAGES OR ITEMS, ETC.

2. ENCODE CLEARLY THE TEST BOOKLET SERIES CODE A, B, C OR D AS THE CASE MAY BE IN THE APPROPRIATE PLACE IN THE ANSWER SHEET.

3. You have to enter your Roll Number on the Test Booklet in the provided alongside this line. DO NOT write anything else on the Test Booklet.

4. This Test Booklet contains 100 questions. Each question is printed in English. Each item comprises four responses (answers). You will select the response which you want to mark on the Answer Sheet. In case you feel that there is more than one correct response, mark the response which you consider the best. In any case, choose ONLY ONE response for each item.

5. All items carry equal marks.

6. Penalty for wrong answers:

THERE WILL BE PENALTY FOR WRONG ANSWERS MARKED BY A CANDIDATE IN THE OBJECTIVE TYPE QUESTION PAPERS.

- I. There are four alternatives for the answer to every question. For each question for which a wrong answer has been given by the candidate, One-Third of the marks assigned to that question will be deducted as penalty.
- II. If a candidate gives more than one answer, it will be treated as a wrong answer even if one of the given answers happens to be correct and there will be same penalty as above to that question.
- III. If a question is left blank, i.e., no answer is given by the candidate, there will be no penalty for that question.

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TOPICS COVERD: Space Technology, Defence, Electronics and Green Tech, Information Technology (IT)+Robotics(A.I), Nuclear + Particle Science (Nano Technology), Current Affairs September 2021



Q.1) Consider the following statements with reference to the Defence Trade and Technology Initiative (DTTI) :

1. The aim was to strengthen the US and India's defence industrial base by moving away from the traditional "buyer-seller" dynamic toward a more collaborative approach
2. Under this initiative both countries had signed agreements on GSOMIA, LEMOA and COMCASA.

Which of the above statements is/are correct ?

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.2) Consider the following statements with reference to the SCO Exercise Peaceful Mission 2021 :

1. The 6th Edition of SCO Exercise Peaceful Mission 2021 hosted by China started in the Orenburg Region of South West China.
2. The Exercise Peaceful Mission 2021 is based on joint counter terrorism operations at operational and tactical level in an urban environment.

Which of the above statements is/are correct ?

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.3) Consider the following statements with reference to the Shukrayaan Mission:

1. The Shukrayaan orbiter will be the second mission to Venus by the India Space Research Organization (ISRO).
2. It aims to investigate the surface processes and shallow subsurface stratigraphy; and solar wind interaction with Venusian Ionosphere.

Which of the above statements is/are correct ?

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.4) Consider the following statements with reference to the New Space India Limited:

1. New Space India Limited is a Central Public Sector Enterprise of the Government under the administrative control of the Department of Space.
2. New Space India Limited will handle ISRO's commercial deals for satellites and launch vehicles with foreign customers.

Which of the above statements is/are correct ?

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.5) Which of the following statements is incorrect with respect to the Spacecom Policy 2020?

- a) The Indian Space Research Organisation (ISRO) has released the draft of a new Spacecom Policy 2020.
- b) It governs the commercial use of orbital slots, satellites, and ground stations for communication needs.
- c) The policy will allow only Indian entities to seek authorisation for orbital slots for new satellites, services based on existing satellites and setting up new ground stations.
- d) None of the above

Q.6) Consider the following statements with reference to the Solar eclipses :

1. Solar eclipses happen when the Earth moves directly between the sun and Moon.
2. The ring of fire eclipse occurs When earth is away from the moon, resulting in a narrow band of light around the dark colour of the Earth.

Which of the above statements is/are correct ?

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.7) Consider the following statements with reference to the Black hole :



1. A black hole is a place in space where gravity pulls so much that even light cannot get out as gravity is so strong because matter has been squeezed into a tiny space.
2. Polarised light waves oscillate in only one direction While, unpolarised light waves oscillate in many different directions resulting in the presence of strong magnetic fields around the black hole.

Which of the above statements is/are correct ?

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.8) What is the purpose of the NASA Perseverance Rover?

- a) To detect neutrinos
- b) To detect gravitational waves
- c) To explore the past habitability of Mars.
- d) To study the effect of solar flares on our communication systems

Q.9) Consider the following statements with reference to the Exoplanet :

1. An exoplanet is any planet beyond our solar system.
2. NASA Transiting Exoplanet Survey Satellite (TESS) mission employs the transit method to detect exoplanets.

Which of the above statements is/are correct ?

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.10) Consider the following statements with reference to the Dark Matter :

1. Dark matter, a component of the universe whose presence is discerned from its gravitational attraction rather than its luminosity.
2. Dark matter interacts with the electromagnetic force to absorb, reflect or emit light, making it extremely hard

to spot.

Which of the above statements is/are correct ?

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.11) The terms Phobos and Deimos are sometimes seen in the news in the context of -

- a) Observation and understanding of the Universe
- b) Study of the solar and the lunar eclipses
- c) moons of the Mars
- d) Origin and evolution of living organisms on the Earth

Q.12) What is the purpose of the Large Hadron Collider (LHC) ?

- a) To study the effect of climate change
- b) To study the dark matter
- c) To detect the effectiveness of missile defence system
- d) To study the effect of solar flares on our communication systems

Q.13) What is the purpose of the Advanced Telescope for High energy Astrophysics project?

- a) To study the black hole.
- b) To study the origin and evolution of living organisms.
- c) To detect the effectiveness of missile defence system
- d) To study the effect of solar flares on our communication systems

Q.14) Consider the following statements with reference to PSLV and GSLV:

1. PSLV has the capability to put a heavier payload in orbit than the GSLV.
2. PSLV is designed mainly to deliver earth observation or remote sensing satellites, whereas GSLV has been



designed for launching communication satellites.

Which of the above statements is/are correct ?

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.15) Consider the following statements with reference to the Sun synchronous and Geosynchronous orbit:

1. Sun synchronous orbit is located at the altitude of 35000 Km and Geosynchronous orbit is located at altitudes of 700 to 800 Km.
2. Sun synchronous orbit satellites help in monitoring activities near the poles unlike geosynchronous orbit satellites.

Which of the above statements is/are correct ?

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.16) What is the purpose of LIGO Observatory ?

- a) To detect neutrinos
- b) To detect gravitational waves
- c) To detect the effectiveness of missile defence system
- d) To study the effect of solar flares on our communication systems

Q.17) Recently launched RESOURCESAT-2A, CARTOSAT-2B, RISAT-2 and OCEANSAT-2 are Which type of satellites?

- a) Remote Sensing satellites
- b) Communication satellite
- c) Both a and b
- d) None of the above

Q.18) Consider the following statements with reference to Global fire power Index :

1. The index is published by the United Nation Security

Council.

2. The United States military was ranked the most powerful armed force in the world closely followed by Russia, China and India.

Which of the above statements is/are correct ?

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.19) Consider the following statements with reference to the Defence India StartUp Challenge:

It aimed at supporting Startups/MSMEs/Innovators to create prototypes and/or commercialize products/solutions in the area of National Defence and Security.

It has been launched by the Defence Ministry in partnership with Atal Innovation Mission.

Which of the above statements is/are correct ?

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.20) Consider the following statements with reference to the INS Vikrant:

1. It is Indigenous Aircraft Carrier likely to be commissioned in 2022.
2. It will have an air component of 30 aircraft, with a top speed of 30 knots (approximately 55 kmph)

Which of the above statements is/are correct ?

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.21) Which of the following is not an objective of the Defence Production and Export Promotion Policy 2020 ?

- a) To promote export of defence products and become part of the global defence value chains.



- b) To Promote import of defence products and become part of the global defence value chains.
- c) To reduce dependence on imports and take forward "Make in India" initiatives through domestic design and development.
- d) None of the above

Q.22) US-2 Amphibious Aircraft is developed by which of the following countries?

- a) United State of America
- b) China
- c) Japan
- d) Russia

Q.23) Consider the following statements with reference to the Chaff technology :

1. Chaff is a critical defence technology used to protect fighter aircraft from hostile radar threats.
2. This technology is developed by the Defence Research and Development Organisation (DRDO).

Which of the above statements is/are correct ?

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.24) With reference to the Essential Defence Services bill 2021, which of the following statements is/are correct?

1. It prohibits any agitation and strike by anyone engaged in the essential defence services.
2. All offences punishable will be non-cognisable and non-bailable.
3. It will amend the Industrial Disputes Act, 1947 to include essential defence services under public utility services.

Select the correct answer using the code given below:

- a) 1 only
- b) 1 and 3 only
- c) 3 only
- d) 1, 2 and 3

Q.25) Which of the following statements is incorrect with respect to the Man-Portable Anti-Tank Guided Missile (MPATGM) ?

- a) It is a low weight and fire & forget missile developed by Israel.
- b) The missile is incorporated with state-of-the-art Miniaturized Infrared Imaging Seeker along with advanced avionics.
- c) The man-portable missile is launched using a tripod designed for a maximum range of 2.5 km with a launch weight of less than 15 Kg.
- d) It is developed by the Defence Research & Development Organisation (DRDO).

Q.26) Consider the following statements with reference to the titanium alloy:

1. The titanium alloys are unique due to their ductility, higher strength, fracture toughness and fatigue.
2. It is used in aircraft, armor plating, naval ships, spacecraft, and missiles.

Which of the above statements is/are correct ?

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.27) With reference to the Drone Rules, 2021, which of the following statements is/are correct?

1. The aim of the Rules is to create a 'digital sky platform which is a business-friendly single-window online system
2. If the weight is more than 1000 kilograms, the provisions of the Aircraft Rules, 1937 shall apply instead of the Drone Rules.
3. The Rules prohibit the operation of a drone in a red zone or yellow zone without prior permission. However, no such provision is required in the Green Zone.

Select the correct answer using the code given below:

- a) 1 only
- b) 1 and 3 only
- c) 3 only
- d) 1, 2 and 3



Q.28) Recently India received 24 MH-60 Romeo multirole helicopters from which of the following countries?

- a) USA
- b) China
- c) Russia
- d) Israel

Q.29) Consider the following statements with reference to the single crystal blade technology:

1. Single crystal blade technology has been developed by the Defence Metallurgical Research Laboratory (DMRL)
2. Single Crystal Blade Technology is a Grain Boundary technology used in making blades to overcome the problem of creep in helicopters.

Which of the above statements is/are correct ?

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.30) Consider the following statements with reference to the Rudram missile :

1. It is India's first indigenous anti-radiation missile which is generally part of their air defence systems.
2. Rudram is an air-to-air missile, designed and developed by the Defence Research and Development Organisation (DRDO).

Which of the above statements is/are correct ?

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.31) Recently India had signed an agreement of S-400 Triumf SA-21 Growler Air Defence System with which of the following countries?

- a) USA
- b) China
- c) Russia

- d) Israel

Q.32) Consider the following statements with reference to the Defence Acquisition Council:

1. The Defence Acquisition Council is the highest decision-making body for deciding on new policies and capital acquisitions for the Army, Navy, Air Force and the Indian Coast Guard.
2. The Prime Minister is the Chairman of the Council.

Which of the above statements is/are correct ?

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.33) Which of the following is/are an application of "Graphene" ?

1. The thermal and electrical conductivity is very high and it can be used as a flexible conductor
2. It can be used in touch screens, light panels and solar cells, where it can replace the rather fragile and expensive Indium-Tin-Oxide (ITO)
3. Graphene can possibly contribute to more accurate resistance standards in metrology.
4. New types of composite materials based on graphene with great strength and low weight can be used in satellites and aircraft.

Select the correct answer using the codes given below:

- a) 1, 2 and 4 only
- b) 1 and 2 only
- c) 1 and 3 only
- d) All the above

Q.34) "Enormous use of nanoparticles (NPs) considered may pose risks to the environment and biological systems" in this context, Consider the following statements about potential hazards of Nanoparticles?

1. Nanopollution is the reason for lung inflammation and heart problems.
2. Nanoparticles used to carry drugs may be toxic to the brain



3. Persistent insoluble nanoparticles may cause problems in the environment

Which of the statements given above is/are correct?

- a) 1 only
- b) 1 and 2 only
- c) 2 and 3 only
- d) all the above

Q.35) With reference to the "Nuclear power" in India , consider the following Statements :

- 1. Nuclear power in India is the fifth-highest generator of electricity in the country.
- 2. India is outside the Nuclear Non-Proliferation Treaty Because of its weapons programme.

Which of the statements given above is/are correct?

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.36) With reference to " Global Initiative to Combat Nuclear Terrorism (GICNT) " , consider the following Statements:

- 1. The GICNT is an international partnership to improve international capacity for prevention, detection and response to nuclear terrorism.
- 2. If a State desires to become a partner in the GICNT it must send a written letter of endorsement of the Statement of Principles to the GICNT
- 3. Currently, the International Atomic Energy Agency (IAEA) is the only official observer of the GICNT.

Which of the statements given above is/are correct?

- a) 1 only
- b) 1 and 2 only
- c) 2 and 3 only
- d) 1 and 3 only

Q.37) Beyond electricity generation, there are many applications that can use nuclear power. Which of the following is /are possible applications of nuclear power ?

- 1. seawater desalination

- 2. hydrogen production
- 3. heating for industry such as glass and cement manufacturing, metal production
- 4. refining and synthesis gas production

Which of the statements given above is/are correct?

- a) 1 only
- b) 1 and 2,3 only
- c) 2 and 4 only
- d) all the above

Q.38) Which of the following statements is not correct regarding "The Large Hadron Collider (LHC)"?

- 1. It is a project of NASA for Nuclear Research with collaboration with scientists and hundreds of universities and laboratories, other countries.
- 2. The LHC's goal is to allow physicists to test the predictions of different theories of particle physics.

Select the answer using the code given below:

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.39) Consider the following statements about the "India-based Neutrino Observatory " , proposed to be set up in Theni district in Tamil Nadu,;

- 1. Neutrinos have a very tiny mass, positive charge particles and spin half.
- 2. They can also be produced in the lab.
- 3. The India-based Neutrino Observatory (INO) will study atmospheric neutrinos only.
- 4. Solar neutrinos have much lower energy than the detector can detect.

Select the correct answer using the code given below.

- a) 1 and 3,4 only
- b) 1 and 2 only
- c) 2 and 3 , 4 only
- d) 1, 2 and 3

Q.40) Consider the following statements:

- 1. Information about properties of the sun
- 2. In astronomy to discover the composition of the



universe.

3. Medical Imaging
4. A way to monitor nuclear proliferation
5. Faster global communication.

Which of the above can be possible future applications of "neutrino science" ?

- a) 1 and 5 only
- b) 1, 3, 5 only
- c) 2, 3, 4 only
- d) All the above

Q.41) In the context of "India's nuclear doctrine", Consider the following statements :

1. The Nuclear Command Authority comprises a Political Council and an executive Council.
2. The Political Council is chaired by the National Security Advisor.
3. The Political Council is the sole body which can authorise the use of nuclear weapons.
4. The Executive Council is chaired by the Prime Minister.

Which of the statements given above is/are correct ?

- a) 1 only
- b) 1 and 2 only
- c) 1 and 3 only
- d) 1, 2 and 3

Q.42) " Flux Trapping Effect "often seen in news related to -

- a) It is a phenomenon about the earth's magnetic cycle
- b) It occurs in ecosystems where a very few resources are remaining
- c) Property of superconductors where a rare earth magnet floats above a cooled superconductor.
- d) It is found in the ocean due to excessive plastic accumulation

Q.43) Consider the following events:

1. Establishment of The Atomic Energy Commission of India
2. Indian Institute of Technology Kharagpur (IIT

Kharagpur) established.

3. Formation of The Defence Research and Development Organisation (DRDO)
4. National Physical Laboratory of India established.

Which of the following is the correct chronological sequence of the above events from the earliest?

- a) 3-2-1-4
- b) 1-3-4-2
- c) 4-1-2-3
- d) 2-1-3-4

Q.44) With reference to "Proton therapy for cancer", consider the following Statements :

1. It is a type of radiation therapy in a treatment that uses high-energy beams to treat tumors.
2. Proton therapy can be utilised for young adults and teenagers to reduce the risk of second cancer.
3. The use of pencil-beam technology that provides a high degree of precision.

Which of the statements given above is/are correct ?

- a) 1 and 2 only
- b) 2 only
- c) 1 and 3 only
- d) All the above

Q.45) Consider the following Statements:

1. Superconducting magnets can be used for accelerating the particles in the Large Hadron Collider.
2. SQUIDS (superconducting quantum interference devices) can be used in the production of highly sensitive magnetometers.
3. Superconductors Can be used for the development of high-intensity Electromagnetic Impulse (EMP).

Which of the above can be possible applications of "Superconductivity" ?

- a) 1 only
- b) 1 and 2 only
- c) 1 and 3 only
- d) All the above



Q.46) The "Technical Cooperation Report " is released by which of the following organizations ?

- a) World Bank
- b) International monetary fund
- c) United Nation Development Program
- d) International Atomic Energy Agency (IAEA)

Q.47) With reference to the "Nuclear Suppliers Group (NSG) " which of the following Statements is/are correct?

- 1. It aims to control the proliferation of nuclear weapons by curbing the export of nuclear weapons development materials and related technology.
- 2. India has been a member of the Nuclear Suppliers Group since 2007 .

Select the correct answer using the code given below :

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.48) For which of the following reasons should India join NSG and it will be beneficial for India?

- 1. Provide the country with access to foreign-sourced nuclear material and equipment.
- 2. Increasing the business of these nuclear materials will enable India to make better versions of nuclear breeders and export them to smaller countries.
- 3. May get an opportunity to initiate talks about the plutonium trade for its thorium program and gain massive domestic profits.

Select the correct answer using the code given below :

- a) 1 and 2 only
- b) 2 only
- c) 1 and 3 only
- d) 1 ,2 ,3 only

Q.49) With reference to "The Non-Proliferation Treaty (NPT)", consider the following Statements:

- 1. The NPT is a multilateral treaty aimed at limiting the spread of nuclear weapons including three elements: non-proliferation, disarmament, and peaceful use of

nuclear energy.

- 2. All United Nations countries signed the Non-Proliferation Treaty (NPT).

Which of the statements given above is/are correct ?

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.50) India's first Lithium refinery set up in which of the following states of India?

- a) Gujarat
- b) Maharashtra
- c) Telangana
- d) Kerala

Q.51) Which of the following is/are the benefit of Offshore wind energy production?

- 1. Absence of any obstruction.
- 2. Wind speed is high and is consistent in direction.
- 3. Offshore wind farms have a higher 'capacity utilisation factor' CUF than onshore wind farms.
- 4. A huge amount of land is required for its installation.
- 5. Maintenance and installation of Offshore wind turbines is more costly than onshore wind turbines.

Select the correct answer using the codes given below:

- a) Only 1, 2, 3 and 4
- b) Only 1, 2, and 3
- c) Only 1, 2, 4 and 5
- d) All of the above

Q.52) Which of the following is/are positive implications of the use of solar energy Technologies?

- 1. Electricity generation.
- 2. Production of thermal energy.
- 3. Reduction of emission of greenhouse gas.
- 4. Increased regional/national energy independence.

Select the correct answer using the codes given below:

- a) Only 1, 2 and 3
- b) Only 1, 3 and 4
- c) Only 1 and 3



d) 1, 2, 3 and 4

Q.53) Consider the following statements regarding solar energy Technologies:

1. Photovoltaic solar systems directly convert the solar energy to electricity through photo-electric effect.
2. Concentrating solar power, which converts the solar radiation into heat.

Which of the statements given above is/are correct ?

- a) Only 1
- b) Only 2
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.54) Which of the following are the main reasons for the use of silicon in solar panels?

1. High quality semiconductor.
2. Available in abundance on earth.
3. Low weight volume ratio.
4. Extended life cycle.
5. Robustness and Strength.
6. Easy to produce at high cost.

Select the correct answer using the codes given below:

- a) Only 1, 2, 3, 4, and 5
- b) Only 1, 4, 5 and 6
- c) Only 1, 3, 4 and 5
- d) All of the above

Q.55) Which of the following is/are technological solutions to fight climate change?

1. Stratospheric aerosol injection.
2. Marine cloud brightening.
3. Ground-based albedo modification.

Select the correct answer using the codes given below:

- a) 1, 2 and 3
- b) 1 and 2 Only
- c) 2 and 3 Only
- d) 1 and 3 Only

Q.56) Which of the following techniques is/are related to 'Carbon Dioxide Removal' ?

1. Carbon capture and storage (CCS)
2. Ocean fertilisation
3. Soil carbon sequestration
4. Deforestation

Select the correct answer using the codes given below

- a) 1, 2 and 3 Only
- b) 2, 3 and 4 Only
- c) 1, 3 and 4 Only
- d) 1, 2 and 4 Only

Q.57) Match the following Pair:

Target - Type of Renewable Energy

1. 100 GW - Solar energy
2. 60 GW - Wind energy
3. 20 GW - Bio-power
4. 15 GW - Small Hydro-power

Which of the following Pair is/are correct regarding the India's target of installation of renewable energy by 2022:

- a) Only 1 and 2
- b) Only 1, 2 and 3
- c) Only 1, 2 and 4
- d) All of the above

Q.58) Which of the following statements is/are correct with regard to Eligibility of becoming a member of the international solar Alliance?

1. All solar rich countries which live fully between the Tropic of Cancer and Tropic of Capricorn.
2. All the member countries of the United Nations.

Select the correct answer using the codes given below:

- a) Only 1
- b) Only 2
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.59) Consider the following statements regarding Quantum Computer:

1. Quantum computers have the ability to process so much more information and so much faster compared to classical computers.



2. In a Quantum computer information is expressed through qubit, which is described as a 0 and 1 at the same time giving us more processing power.
3. Superposition and Entanglement are what give quantum computers the ability to process so much more information.

Which of the above statements is/are correct ?

- a) Only 1 and 2
- b) Only 2 and 3
- c) Only 3 and 1
- d) All of the above.

Q.60) Which of the following statements is correct regarding 'Sycamore'?

- a) It is a Quantum computer of Google Inc.
- b) It is a supercomputer launched by China.
- c) It is a type of Malware which Steals financial data of victims.
- d) None of the above.

Q.61) Which of the following statements is incorrect regarding "DigiLocker"?

- a) It was launched by the 'Ministry of Electronics and information technology' in 2015.
- b) Legacy documents can be electronically signed by citizens using the eSign facility.
- c) DigiLocker account users get a dedicated cloud storage space that is linked to their Aadhaar (UIDAI) number.
- d) Citizens can access their digital documents anytime, anywhere but can't share it online.

Q.62) Which of the following statements is/are correct regarding 'PARAM-Siddhi' ?

1. It is the high performance computing artificial intelligence supercomputer.
2. It is established under the 'National Supercomputing Mission'.
3. It has achieved a global ranking of 62 in TOP-500 most powerful non-distributed computer systems in the world.

Select the correct answer using the codes given below:

- a) Only 1
- b) Only 1 and 3
- c) Only 2
- d) 1, 2, and 3

Q.63) Consider the following statements regarding "National Supercomputing Mission":

It is being jointly steered by the Department of Science and Technology (DST) and the Department of Electronics and Information Technology (DeitY). It is implemented by the Centre for Development of Advanced Computing (C-DAC), Pune, and the Indian Institute of Science (IISc), Bengaluru.

Which of the statements given above is/are correct ?

- a) Only 1
- b) Only 2
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.64) Which of the following is/are applications of Supercomputers ?

1. Climate Modelling and Weather Prediction
2. Aerospace Engineering
3. Computational Biology
4. Atomic Energy Simulations
5. National Security
6. Seismic Analysis

Select the Correct answer using the codes given below:

- a) Only 1, 2, 3, 4 and 6
- b) Only 1, 3, 4, 5 and 6
- c) Only 1, 2, 4, 5 and 6
- d) All of the above.

Q.65) Which of the following is/are the latest Mass Rapid Transportation Technologies approved by NITI Aayog?

1. Hyperloop
2. metrino
3. Stadler buses
4. Pod taxis
5. Metro trains

Select the correct answer using the codes given below:



- (a) Only 1, 2, 3 and 4
- (b) Only 2, 3, 4 and 5
- (c) Only 1, 3, 4 and 5
- (d) All of the above.

- a) Only 1, 2, 3 and 4
- b) Only 2, 3, 4, 5 and 6
- c) Only 1, 3, 4, 5 and 6
- d) All of the above

Q.66) Which of the following statement/s is/are Correct regarding the recently approved "Drone rules, 2021" ?

1. No remote pilot licence required for micro drones and nano drones at any circumstances.
2. Coverage of drones increased from 300 kg to 500 kg.

Select the correct answer using the codes given below:

1. Only 1
2. Only 2
3. Both 1 and 2
4. Neither 1 nor 2

Q.67) Which of the following pairs is/are correct regarding operating zones of drones in India?

- | Operating Zones | Regulations/Permission |
|---------------------|--|
| Green Zones | - no permission required. |
| Yellow Zones | - controlled airspace and permission required from the concerned state government. |
| Red Zone | - drones operations permitted under exceptional circumstances and permission required from the central government. |

Select the correct answer using the codes given below:

- a) Only 1 and 2
- b) Only 2 and 3
- c) Only 1 and 3
- d) 1, 2 and 3

Q.68) Which of the following is/are applications of Drones?

1. Soil Health Assessment
2. Mining Field
3. Surveillance
4. Emergency Response
5. Transportation- Traffic Management
6. Geo-spatial Mapping

Select the correct answer using the codes given below:

Q.69) Which of the following statements is correct regarding "Pegasus" which is recently in the news?

- a) It is Israel based spyware software.
- b) It is China based Malware which steals the personal financial data of victims.
- c) It is a type of facial recognition Technology.
- d) None of the above is correct.

Q.70) Consider the following pair:

Virtual Reality (VR) - Immerses users in a fully artificial digital environment.

Augmented Reality (AR) - Overlays virtual objects in the real-world environment.

Mixed Reality (MR) - Not just overlays but placing virtual objects to the real world.

Which of the pairs given above is/are correct ?

- a) 1 only
- b) 2 only
- c) 3 only
- d) All of the above

Q.71) Which of the following statements correctly define "Metaverse" which is recently in the news?

- a) An online world where people can exist and communicate in shared virtual spaces.
- b) It is a new Virtual Reality headset, which was launched by Google.
- c) Augmented Reality based computer game for children.
- d) None of the above is correct.

Q.72) Which of the following statements is Incorrect regarding 'BharatNet Project' ?

1. It comes under the Ministry of Communication and Information Technology.
2. It was funded by the 'Universal Obligation Service



Fund'.

Select the correct answer using the codes given below:

- a) Only 1
- b) Only 2
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.73) Which of the following statements is/are correct regarding "Dark Fibre"?

It is also known as Unlit Fibre which is currently not in use in fibre optic communication.

It will help the telecom service providers to cut down on their capital expenditure.

Select the correct answer using the codes given below:

- a) Only 1
- b) Only 2
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.74) Consider the following statements with reference to the Index of Industrial Production :

1. Index of Industrial Production data or IIP as it is commonly called is an index that tracks service sector activity in an economy.
2. Mining, manufacturing, and electricity are the three broad sectors in which IIP constituents

Which of the above statements is/are correct ?

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.75) Which of the following statements is incorrect with respect to the Hydrogen Fuel Cell ?

- a) In a hydrogen fuel cell, it generates electricity with only water and heat as byproducts of the process.
- b) Unlike electric vehicles, FCVs are zero-emission vehicles.
- c) National Hydrogen Mission which aims for generation of hydrogen from green power resources.

- d) None of the above

Q.76) The term Stablecoin is used in the context of ?

- a) Paper Currency
- b) Crypto Currency
- c) Coinage
- d) None of the above

Q.77) The Global Innovation Index is published by which of the following organizations ?

- a) World Bank
- b) Organisation for Economic Co-operation and Development
- c) World Intellectual Property Organization
- d) None of the above

Q.78) Consider the following statements with reference to the National Single Window System (NSWS).

1. It offers a single dashboard, allowing businesses to apply for clearances, track progress and respond to queries
2. It is developed by the Department for Promotion of Industry and Internal Trade (DPIIT) along with Invest India.

Which of the above statements is/are correct ?

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.79) Which of the following statements is incorrect with respect to the State of Climate Services report 2021?

- a) The report is released by World Meteorological Organization (WMO)
- b) It focuses on Terrestrial Water Storage.
- c) The biggest losses of Terrestrial Water Storage have occurred in Antarctica and Greenland.



- d) In India the Terrestrial Water Storage has been increased at a rate of at least 5 cm per year.

- c) Dara Shikoh
d) Jahangir

Q.80) Who among the following is associated with the philosophy of Total Revolution?

- a) Bal Gangadhar Tilak
b) Jawaharlal Nehru
c) Mohandas Karamchand Gandhi
d) Jay Prakash Narayan

Q.81) Which of the following statements is incorrect with respect to 'International Energy Agency' ?

- a) The International Energy Agency is a Paris-based autonomous nongovernmental organisation.
b) It was established under the framework of the Organisation for Economic Co-operation and Development (OECD) in 1974 in the wake of the 1973 oil crisis.
c) It is best known for the publication of its annual World Energy Outlook.
d) India became an Associate member of the International Energy Agency.

Q.82) Which of the following statements is incorrect with respect to 'plastic waste recycling rules' ?

- a) Plastic packaging, as per the rules divided into three categories.
b) Producers of plastic would be obliged to declare to the government how much plastic they produce annually.
c) If entities cannot fulfil their obligations, they would on a "case basis by case" be permitted to buy certificates making up for their shortfall.
d) Non-compliance would invite a traditional fine instead of environmental compensation.

Q.83) Who among the following Mughal Emperors associated with the most important works Majma-ul-Bahrain and Sirr-i-Akbar?

- a) Aurangzeb
b) Akbar

Q.84) Consider the following statements with reference to the leaded petrol :

1. The campaign against lead petrol was led and supported by the UNEP and its Partnership for Clean Fuels and Vehicles (PCFV)
2. The use of lead as a petrol additive has been a catastrophe for public health.

Which of the above statements is/are correct ?

- a) 1 only
b) 2 only
c) Both 1 and 2
d) Neither 1 nor 2

Q.85) Which of the following statements is incorrect with respect to the Universal Postal Union ?

- a) It was established in 1874 with headquarters in the Swiss capital Bern.
b) It is the first oldest international organization worldwide.
c) Universal Postal Union is the primary forum for cooperation between postal sector players.
d) None of the above

Q.86) Which of the following statements is incorrect with respect to the West Nile virus (WNV) ?

- a) It is a member of the flavivirus genus.
b) It is the leading cause of mosquito-borne disease.
c) Human-to-human transmission of WNV through casual contact has been documented
d) None of the above.

Q.87) Consider the following statements with reference to the Programme for International Student Assessment (PISA):

1. The Programme for International Student Assessment (PISA) is a biennial international survey which aims to evaluate education systems worldwide.
2. It is coordinated by the Organization for Economic



Cooperation and Development (OECD).

Which of the above statements is/are correct ?

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.88) Consider the following pairs :

Species	IUCN Red list Status
1. Gharial.	- Critically Endangered
2. Mugger Crocodile.	- Least Concern
3. Saltwater Crocodile	- Vulnerable.

Which of the pairs given above is/are correctly matched?

- a) 1 only
- b) 1 and 2
- c) 3 only
- d) 2 and 3

Q.89) Recently, the India-led Resolution 2593 adopted by United Nations Security Council (UNSC) deals with -

- a) Taliban
- b) Climate Change
- c) Terrorism
- d) None of the above

Q.90) Consider the following statements with reference to the Deputy Speaker:

- 1. The Deputy Speaker is elected by the Rajya Sabha from amongst its members.
- 2. The date of election of the Deputy Speaker is fixed by the President.

Which of the above statements is/are correct ?

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.91) Consider the following statements with reference to the Pulicat Lake:

- 1. Pulicat Lake is the second-largest brackish water lake

in the country, next only to Chilika lake.

- 2. It lies at the border of the states of Andhra Pradesh, Karnataka and Tamil Nadu.

Which of the above statements is/are correct ?

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.92) The Durand line was demarcated between -

- a) Afghanistan and China
- b) Afghanistan and Pakistan
- c) India and Pakistan
- d) India and Myanmar

Q.93) Consider the following statements with reference to the Eastern Economic Forum :

- 1. Its aim is to support the economic development of Russia's Far East and to expand international cooperation in the Asia-Pacific region.
- 2. The Forum's business programme includes a number of business dialogues with leading partner countries in the Middle East region.

Which of the above statements is/are correct ?

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.94) Which of the following statements is incorrect with respect to the Inspiration4 ?

- a) Inspiration4 is the world's first all-civilian mission to orbit.
- b) The flight will be operated by the Indian Space Research Organisation (ISRO).
- c) This will be the farthest distance travelled by a crewed mission since 2009.
- d) Inspiration4 is seen as part of an effort to open up space travel to non-professionals.



Q.95) Consider the following statements with reference to the Karbi Anglong Agreement:

1. Karbi Anglong Agreement is a tripartite agreement signed by the Centre, five insurgent groups active in Karbi Anglong, and the Assam government.
2. The Consolidated Fund of India will be amended to meet the resources of the Karbi Anglong Autonomous Council.

Which of the above statements is/are correct ?

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.96) Recently, Astrophysical Journal identifies Hycean worlds a new class of -

- a) Exoplanets
- b) Asteroids
- c) Moon
- d) None of the above

Q.97) Consider the following statements with reference to the 13th BRICS Summit :

1. It will be the second time that India will be hosting the BRICS Summit after 2016.
2. They deliberate on important issues under the three pillars of political and security, economic and financial and cultural and people to people exchanges.

Which of the above statements is/are correct ?

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.98) Consider the following statements with reference to the Dugong:

1. The dugong, also called the sea cow, is a carnivorous mammal.
2. India's first dugong conservation reserve is set up by the Tamil Nadu State government in the Palk Bay.

Which of the above statements is/are correct ?

- a) 1 only

- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.99) Consider the following statements with reference to the National Institute Ranking Framework:

1. National Institute Ranking Framework is the first-ever effort by the government to rank higher education institutions (HEIs) in the country.
2. It ranks only Indian higher education institutions (HEIs).

Which of the above statements is/are correct ?

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.100) Which of the following statements is incorrect with respect to Pradhan Mantri Gramin Digital Saksharta Abhiyan ?

- a) It is expected to be one of the largest digital literacy programmes in the world.
- b) Scheme is applicable to both urban and rural areas of the country.
- c) Beneficiaries should be Digitally Illiterate, who are only from rural areas of the country.
- d) Preference would be given to SC, ST, BPL, women, differently-abled persons and minorities.



Q.1) Ans: c

Exp:

- **Statement 1 is correct:** The **Defence Trade and Technology Initiative (DTTI)** was formed in 2012 between the **US and India**, to enhance the bilateral relations in defence by venturing into the field of **advanced defence research and development and manufacturing**. The aim was to **strengthen the US and India's defence industrial base** by moving away from the traditional "buyer-seller" dynamic toward a more collaborative approach. This would be through exploring new areas of technological collaboration through co-development and co-production. Co-development of a defence product between two nations may not be a new concept, but formalising the process to execute multiple projects of co-development is an ambitious innovation.
- **Statement 2 is correct:** The **enabling agreements** which have been signed between the two countries are **GSOMIA, LEMOA and COMCASA**. These agreements along with BECA, which is slated to be signed in the forthcoming **2+2 ministerial dialogue** enhance cooperation and interoperability between the armed forces of the two democracies. All these steps provide greater avenues to identify and execute meaningful projects under DTTI.

Source

: <https://www.financialexpress.com/defence/india-us-defence-trade-and-technology-initiative-is-the-silent-enabler/2083679/>

Q.2) Ans: b

Exp:

- **Statement 1 is incorrect:** The 6th Edition of SCO Exercise Peaceful Mission 2021 hosted by **Russia started in the Orenburg Region of South West Russia**. The aim of the exercise is to foster close relations between SCO Member States and to enhance abilities of the military leaders to

command multinational military contingents. An Indian military contingent consisting of an all arms combined force of 200 personnel from Indian Army and Indian Air force is participating in the exercise.

- **Statement 2 is correct:** The Exercise Peaceful Mission 2021 is based on joint counter terrorism operations at operational and tactical level in an urban environment in which Armies and Air Forces of all SCO member states are participating. The troops will train, share and rehearse tactical drills which will culminate in a final validation exercise, where-in troops from all Armies and Air Forces will jointly undertake operations in a controlled and simulated environment.

Source

: <https://pib.gov.in/PressReleaseIframePage.aspx?PRID=1756433>

Q.3) Ans: b

Exp:

- **Statement 1 is incorrect:** The Shukrayaan orbiter will be the first mission to Venus by the India Space Research Organization (ISRO) and will study the planet for four years. Shukrayaan is set to launch on India's GSLV Mk II rocket, but it may go on the more powerful GSLV Mk III rocket to carry more instruments or fuel.
- **Statement 2 is correct:** Shukrayaan will also bring an instrument to Venus to examine the planet's atmosphere in infrared, ultraviolet and submillimeter wavelengths. Scientists announced the possible detection of phosphine — a life-friendly element — in Venus' atmosphere, although many in the science community remain skeptical of the findings



customers. NSIL will deal with capacity building of local industry for space manufacturing

Source : <https://www.nsilindia.co.in/>

Q.5) Ans: d

Exp:

Spacecom Policy 2020

- In line with its mandate to open up the space sector for private players, the Indian Space Research Organisation (ISRO) has released the draft of a new Spacecom Policy 2020 to govern the commercial use of orbital slots, satellites, and ground stations for communication needs. The policy details how private players can get authorization for setting up new communication satellites and ground stations.
- Allowing private players in the space communication sector will also enable India to keep pace with the growing demand for satellite-based broadcasting, network connectivity, and global mobile personal communication. This will also establish India as a significant player in the global space communication sector, the document states.
- The policy will allow only Indian entities to seek authorisation for orbital slots for new satellites, services based on existing satellites and setting up new ground stations. However, any company sending a communication satellite in space will also be liable for any damages to other objects in space and the environment. To cover this, companies will have to provide a financial guarantee or insurance cover at the time of seeking authorisation from the department of space and later from Indian National Space Promotion and Authorisation Centre (IN-SPACe).
- In addition, all existing ISRO -operated satellites—such as INSAT and GSAT—will be handed over to the PSUs, like the New Space India Limited, at “no or notional cost” which can then charge as deemed fit for commercialisation.

Extra Edge by Only IAS
About Venus :

Venus is often described as the “twin sister” of the Earth because of the similarities in size, mass, density, bulk composition and gravity. It is believed that both planets share a common origin, forming at the same time out of a condensing nebulosity around 4.5 billion years ago. Venus is around 30 percent closer to the Sun as compared to Earth resulting in much higher solar flux.

Source : <https://www.space.com/india-venus-orbiter-shukrayaan-2024-launch>

Q.4) Ans: a

Exp:

- **Statement 1 is correct:** NSIL is a Central Public Sector Enterprise of the Government of India. It was established in 2019 under the administrative control of the Department of Space. Its Headquarter at Bengaluru. NSIL is the commercial arm of Indian Space Research Organisation (ISRO) with the primary responsibility of enabling Indian industries to take up high technology space related activities and is also responsible for promotion and commercial exploitation of the products and services emanating from the space industry.
- **Statement 2 is incorrect:** NSIL is incorporated to carry forward the industry production of space systems and the ISRO's efforts in realising Polar Satellite Launch Vehicles (PSLV) from industry. NSIL differs from ISRO's existing commercial arm Antrix Corporation. Antrix will handle ISRO's commercial deals for satellites and launch vehicles with foreign



Source: <https://www.hindustantimes.com/india-news/isro-releases-draft-policy-to-regulate-space-communication-by-private-players/story-hcrB1xAKZDFNQdI0y4GatJ.html>

Q.6) Ans: d

Exp:

- Statement 1 is incorrect: Solar eclipses happen when the moon moves directly between the sun and Earth, casting a shadow on our planet and blocking out at least some of the sun's light.

There are three types of solar eclipses:

- o Partial solar eclipse: When the sun, moon and earth are not exactly lined up.
- o Total solar eclipse: When the sun, moon and earth must be in a direct line.
- o Annular solar eclipse: It is a particular type of total solar eclipse. It occurs when the sun, moon and earth are not only in a straight line but also in the same plane.

- **Statement 2 is incorrect:** The moon also must be farther away from the earth, which will allow it to not cover the disc of the sun completely, resulting in a narrow band of light around the dark colour of the moon causing the ring of fire to be visible. Therefore, It is also called the ring of fire eclipse. The distance between the earth and the moon at the moment of the eclipse can dictate the type of eclipse that will take place. The distance between the earth and the moon is always changing due to the egg-shaped elliptical orbit of the moon.

Source : <https://www.space.com/ring-of-fire-annular->

[solar-eclipse-2021-death-star.](#)

Q.7) Ans: c

Exp:

- Statement 1 is correct: A black hole is a place in space where gravity pulls so much that even light cannot get out as gravity is so strong because matter has been squeezed into a tiny space. They are invisible and observed through space telescopes with special tools. Black holes can be big or small, the smallest black holes are as small as just one atom.
- Statement 2 is correct: Now, a new image from Event Horizon Telescope (EHT) shows polarised light, waves oscillating in only one direction produced by matter at the edge of the black hole. While, unpolarised light is made up of light waves oscillating in many different directions. Light becomes polarized when it goes through certain filters like, for instance, the lenses of polarized sunglasses, or when it is emitted in hot regions of space that are magnetized, hence it suggests the presence of strong magnetic fields around the black hole.

Source

: <https://www.nasa.gov/audience/forstudents/k-4/stories/nasa-knows/what-is-a-black-hole-k4.html>

Q.8) Ans: c

Exp:

- The Mars 2020 Perseverance Rover will search for signs of ancient microbial life, which will advance NASA's quest to explore the past habitability of Mars. The rover has a drill to collect core samples of Martian rock and soil, then store them in sealed



tubes for pickup by a future mission that would ferry them back to Earth for detailed analysis. Perseverance will also test technologies to help pave the way for future human exploration of Mars.

- Mission steps: Perseverance will collect rock and soil samples in cigar-sized tubes. The samples will be collected, the canisters will be sealed, and left on the ground. It will land, drive, and collect all samples from the different locations, and return to the lander. These samples will be transferred to the Mars Ascent Vehicle which will meet with an Orbiter. The Orbiter will carry the samples back to Earth.

Source: <https://indianexpress.com/article/explained/explained-why-is-mars-so-interesting-to-scientists-and-to-the-adventurer-7194898/>

Q.9) Ans: c

Exp:

- Statement 1 is correct: An exoplanet is any planet beyond our solar system. The first confirmation of detection of exoplanets occurred in 1992. More than 4,400 exoplanets have been discovered till now. Most orbit other stars, but free-floating exoplanets, called rogue planets, orbit the galactic center and are untethered to any star. Exoplanets are made up of elements similar to those of the planets in our solar system, but their mixes of those elements may differ. Some planets may be dominated by water or ice, while others are dominated by iron or carbon.
- Statement 2 is correct: The count of confirmed planets is in the thousands and rising. NASA Transiting Exoplanet Survey Satellite (TESS) mission led by the Massachusetts Institute of Technology (with seed funding from Google)

will look for planets orbiting the brightest stars in Earth's sky. TESS employs the transit method to detect exoplanets. The transit method of detecting exoplanets looks for dips in the visible light of stars and requires that planets cross in front of stars along our line of sight to them. Repetitive, periodic dips reveal a planet or planets orbiting a star.

Source : <https://exoplanets.nasa.gov/what-is-an-exoplanet/overview/>

Q.10) Ans: a

Exp:

- **Statement 1 is correct:** Dark matter, a component of the universe whose presence is discerned from its gravitational attraction rather than its luminosity. By fitting a theoretical model of the composition of the universe to the combined set of cosmological observations, scientists have come up with the composition that we described above, ~68% dark energy, ~27% dark matter, ~5% normal matter
- **Statement 2 is incorrect:** Unlike normal matter, dark matter does not interact with the electromagnetic force. This means it does not absorb, reflect or emit light, making it extremely hard to spot. In fact, researchers have been able to infer the existence of dark matter only from the gravitational effect it seems to have on visible matter.

Source : <https://home.cern/science/physics/dark-matter>

Q.11) Ans: c

Exp:

About the Mars



- Mars is the fourth planet from the Sun and the second-smallest planet in the Solar System. Mars is about half the size of Earth. As Mars orbits the Sun, it completes one rotation every 24.6 hours, which is very similar to one day on Earth (23.9 hours).
- Mars' axis of rotation is tilted 25 degrees with respect to the plane of its orbit around the Sun. This is similar to Earth, which has an axial tilt of 23.4 degrees.
- Like Earth, Mars has distinct seasons, but they last longer than seasons on Earth since Mars takes longer to orbit the Sun (because it's farther away).
- Martian days are called sols—short for 'solar day'. Mars has a thin atmosphere made up mostly of carbon dioxide, nitrogen and argon gases.
- Mars has no magnetic field till date, but areas of the Martian crust in the southern hemisphere are highly magnetized, indicating traces of a magnetic field. Mars has two small moons, Phobos and Deimos, that may be captured asteroids.

Sources: <https://en.wikipedia.org/wiki/Mars>

Q.12) Ans: b

Exp:

Large Hadron Collider

- The Large Hadron Collider (LHC) is the world's largest and most powerful particle collider and the largest machine in the world. Recently, The European Organisation for Nuclear Research (CERN) has approved an experiment designed to look for light and weakly interacting particles at the LHC.
- It is planning a new experiment to look for particles associated with the mysterious dark matter which makes up about 27% of the universe. This novel experiment will help diversify the physics programme of colliders such as the LHC, and will allow the researchers to address unanswered

questions in particle physics from a different perspective. FASER (or the Forward Search Experiment) will complement CERN's ongoing physics programme & extend its discovery potential to several new particles.

Source: <https://home.cern/science/accelerators/large-hadron-collider>

Q.13) Ans: a

Exp:

Advanced Telescope for High energy Astrophysics project

- Athena will be the largest X-ray observatory ever built, investigating some of the hottest and most energetic phenomena in the cosmos with unprecedented accuracy and depth. It is designed to answer two fundamental questions
- how supermassive black holes at the centre of galaxies form and evolve, and how 'ordinary' matter assembles
- to form the wispy 'cosmic web' that pervades the Universe.
- The Objective of Athena is going to measure several hundreds of thousands of black holes, from relatively nearby to far away, observing the X-ray emission from the million-degree-hot matter in their surroundings.
- Source: <https://sci.esa.int/web/athena>

Q.14) Ans: b

Exp:

- Statement 1 is incorrect: GSLV has the capability to put a heavier payload in orbit than the Polar Satellite Launch Vehicle (PSLV). PSLV can carry satellites up to a total weight of 2000 kg into space



and reach up to an altitude of 600-900 km. GSLV can carry weight up to 5,000 kg and reach up to 36,000 km.

- Statement 2 is correct: PSLV is designed mainly to deliver earth observation or remote sensing satellites, whereas GSLV has been designed for launching communication satellites. GSLV delivers satellites into a higher elliptical orbit, Geosynchronous Transfer Orbit (GTO) and Geosynchronous Earth Orbit (GEO).

Source : <https://www.business-standard.com/about/what-is-gslv>

Q.15) Ans: b

Exp:

- **Statement 1 is incorrect:** An earth satellite orbit in which the orbital plane is near polar and the altitude is such that the satellite passes over all places on earth having the same latitude twice in each orbit at the same local suntime. Orbital period of a sun synchronous satellite is about 100 minutes. These satellites are synchronous with respect to the Earth. This means looking from a fixed point from earth, these satellites appear to be stationary. Geosynchronous orbit period is one sidereal day which is about 23 hours, 56 minutes, and 4 seconds. Sun synchronous orbit is located at the altitude of 700 to 800 Km. Geosynchronous orbit is located at an altitude of 35786 Km.
- **Statement 2 is correct:** Sun synchronous orbit satellites help in monitoring activities near the poles unlike geosynchronous orbit satellites. Geosynchronous satellite can view a particular spot on the Earth's surface continuously unlike sun synchronous satellite. A Sun-synchronous orbit is useful for imaging, reconnaissance satellite, and weather satellites, because every time that the

satellite is overhead, the surface illumination angle on the planet underneath it will be nearly the same. Communications satellites are often given geostationary or close to geostationary orbits so that the satellite antennas that communicate with them do not have to move, but can be pointed permanently at the fixed location in the sky where the satellite appears.

Source : <https://www.rfwireless-world.com/Terminology/difference-between-sun-synchronous-orbit-and-geosynchronous-orbit.html>

Q.16) Ans: b

Exp:

Laser Interferometer Gravitational-wave Observatory (LIGO)

- It is World largest gravitational wave observatory for detecting cosmic gravitational waves and for carrying out experiments.
- It Comprises two enormous laser interferometers located thousands of kilometers apart, each having two L-shaped arms of 4km length.
- Two LIGO detectors are already operational in the U.S, at Livingston and Hanford.
- The Japanese detector, KAGRA, recently joined the international network.

LIGO- India – InDIGO

- LIGO-India project is an Indian Initiative in Gravitational wave observations, expected to be completed by 2025.
- It aims to move one advanced LIGO detector from Hanford to Maharashtra(Hingoli district), India.
- Project is piloted by the Dept. of Atomic Energy(DAE) and dept. of Science and Tech(DST).
- This project will help Indian scientists to be a major player in the emerging research frontier of GW astronomy.



Source: <https://www.ligo.caltech.edu/>

Q.17) Ans: a

Exp:

REMOTE SENSING (Earth Observation) SATELLITES

- Starting with IRS-1A in 1988, ISRO has launched many operational remote sensing satellites.
- They are mostly polar, sun-synchronous satellites in low- earth orbit (LEO) at about 800 km from the earth surface.
- Currently, 13 operational satellites are in Sun-synchronous orbit: RESOURCESAT-1, 2, 2A CARTOSAT-1, 2, 2A, 2B, RISAT-1 and 2, OCEANSAT-2, Megha-Tropiques, SARAL and SCATSAT-1, and 4 in Geo-stationary orbit: INSAT-3D, Kalpana & INSAT 3A, INSAT -3DR.
- They are commonly called as remote sensing satellites as they collect information of any object on Earth through the measurement of radiation of the Sun that is reflected and scattered by objects on the surface of the earth.
- Applications covering agriculture, water resources, urban planning, rural development, mineral prospecting, environment, forestry, ocean resources and disaster management.

Q.18) Ans: b

Exp:

- **Statement 1 is incorrect:** Global Firepower has published the Global Firepower Index (GFP) /Military Strength Ranking 2021. There are currently 138 countries considered for the Global Firepower (GFP) annual defense review (which currently covers the year 2021). Each country is evaluated on a multitude of factors related to a

prolonged offensive or defensive military campaign.

- **Statement 2 is correct:** The finalized Global Firepower ranking utilizes over 50 individual factors to determine a given nation's PowerIndex score with categories ranging from military might and financials to logistical capability and geography. The United States military was ranked the most powerful armed force in the world closely followed by Russia and China. India was ranked fourth in the Index with 542 combat aircraft, 17 submarines, 4,730 tanks, and 37 attack helicopters.

Source : <https://www.globalfirepower.com/>

Q.19) Ans: c

Exp:

- Statement 1 is correct: Defence India StartUp Challenge is aimed at supporting Startups/MSMEs/Innovators to create prototypes and/or commercialize products/solutions in the area of National Defence and Security. The vision of the Challenge is two-fold:
 - Help create functional prototypes of products/technologies relevant for national security (prototyping), and spur fast-moving innovation in the India defence sector;
 - Help new tech products/technologies find a market and early customer (commercialization) in the form of the Indian Defence Establishment.
- Statement 2 is correct: Taking the iDEX initiative further, Defence India Startup Challenge "has been launched by the Defence Ministry in partnership with Atal Innovation Mission. The selected applications will be funded up to Rs. 1.5 crores in the form of equity/other relevant structures. The funds will be disbursed in tranches based on the



milestones decided by a high-powered selection committee. The selected applicants may also be supported in terms of access to defence testing facilities and experts for their product/technology development.

platform/article34981900.ece

Q.21) Ans: b

Exp:

Defence Production and Export Promotion Policy 2020

- The Ministry of Defence (MoD) has formulated a draft Defence Production and Export Promotion Policy 2020 (DPEPP 2020) to provide impetus to self-reliance in defence manufacturing and to position India amongst the leading countries of the world in defence and aerospace sectors.
- The DPEPP 2020 is envisaged as an overarching guiding document of MoD to provide a focused, structured and significant thrust to defence production capabilities of the country for self-reliance and exports.

The policy has laid out following goals and objectives:

To achieve a turnover of Rs 1,75,000 Crores (US\$25Bn) including export of Rs 35,000 Crore (US\$5Bn) including export of Rs 35,000 Crore (US\$5 Billion) in Aerospace and Defence goods and services by 2025. To develop a dynamic, robust and competitive Defence industry, including Aerospace and Naval Shipbuilding industry to cater to the needs of Armed forces with quality products.

To **reduce dependence on imports** and take forward "Make in India" initiatives through domestic design and development.

To promote export of defence products and become part of the global defence value chains. To create an environment that encourages R&D, rewards innovation, creates Indian IP ownership and promotes a robust and self-reliant defence industry.

Source: <https://pib.gov.in/PressReleasePage.aspx?PRID=1643194>

Source: <https://www.makeinindia.defence.gov.in/pages/defence-india-startup-challenge>

Q.20) Ans: c

Exp:

- **Statement 1 is correct:** INS Vikrant, is likely to be commissioned in 2022. It is Indigenous Aircraft Carrier which is a part of Atmanirbhar Bharat. At present, India has only one aircraft carrier, the Russian-origin INS Vikramaditya. Earlier, the Defence Acquisition Council (DAC) approved issuance of Request for Proposal (RFP) for six advanced submarines for the Indian Navy under Project-75I.
- **Statement 2 is correct:** The vessel, to be named Vikrant after the decommissioned maiden carrier of the Navy. It will have an air component of 30 aircraft, comprising MiG-29K fighter jets, Kamov-31 airborne early warning helicopters and the soon-to-be-inducted MH-60R multi-role helicopter, besides the indigenous Advanced Light Helicopters. It is expected to have a top speed of 30 knots (approximately 55 kmph) and is propelled by four gas turbines. Its endurance is 7,500 nautical miles at 18 knots (32 kmph) speed. The shipborne weapons include Barak LR SAM and AK-630, while it has MFSTAR and RAN-40L 3D radars as sensors. The vessel has a Shakti EW (Electronic Warfare) Suite.
- Source : <https://www.thehindu.com/news/national/carrier-will-be-the-navys-most-potent->



Q.22) Ans: c

Exp:

- The ShinMaywa US-2 is a Japanese large short takeoff and landing amphibious aircraft developed and manufactured by seaplane specialist ShinMaywa (formerly Shin Meiwa). It was developed from the earlier Shin Meiwa US-1A seaplane, which was introduced during the 1970s.
- The US-2 is an amphibian aircraft, with a primary role of Search and Rescue (SAR). It is a modern version of an earlier aircraft built by the ShinMaywa company. (Amphibians were widely used in the expanse of the Pacific for maritime reconnaissance and patrol by the US Navy and in smaller numbers by the Imperial Japanese Navy until the end of WWII).
- The US-2 has a better operating envelope with a claimed ability to land and take off in waves of up to 3m height.

Source: https://www.shinmaywa.co.jp/aircraft/english/us2/us2_capability.html

Q.23) Ans: c

Exp:

- **Statement 1 is correct:** A chaff is primarily an electronic counter-measure technology used by militaries worldwide to protect high-value targets such as fighter jets or naval ships from radars and radio frequency (RF) guiding mechanisms of the enemy missiles.
 - The chaff deployed in the air reflect multiple targets for the missile guidance systems, thus misleading the enemy radars or deflecting adversary missiles. To ensure survivability of aircraft, CounterMeasure Dispensing System (CMDS) is used which provides passive jamming against infra-red and radar threats.

- Chaff is a critical defence technology used to protect fighter aircraft from hostile radar threats. The importance of this technology lies in the fact that very less quantity of chaff material deployed in the air acts as a decoy to deflect enemy's missiles for ensuring safety of the fighter aircraft.

- **Statement 2 is correct:** Pune and Jodhpur-based facilities of the Defence Research and Development Organisation (DRDO) have jointly developed an advanced chaff technology to safeguard fighter aircraft of the Indian Air Force (IAF) from enemy radar threats. Defence Laboratory, Jodhpur, and High Energy Materials Research Laboratory (HEMRL), Pune, have developed the chaff cartridge meeting qualitative requirements of IAF.

Source : <https://indianexpress.com/article/cities/pune/drdo-develops-advanced-chaff-technology-to-protect-iaf-jets-against-radar-threats-7461515/>

Q.24) Ans: b

Exp:

- **Statement 1 is correct:** It seeks to replace the ordinance issued in June 2021 and prohibits any agitation and strike by anyone engaged in the essential defence services.

Essential	Defence	Services:
o It includes any service in any establishment or undertaking dealing with production of goods or equipment required for defence related purposes or any establishment of the armed forces or connected with them or defence.		
o It also includes services that, if ceased, would affect the safety of the establishment engaged in such services or its employees.		
o In addition, the government may declare any service		



as an essential defence service if its cessation would affect the:

- o Production of defence equipment or goods.
- o Operation or maintenance of industrial establishments or units engaged in such production.
- o Repair or maintenance of products connected with defence.

- **Statement 2 is incorrect:** It is defined as cessation of work by a body of persons acting together, which includes:

Mass casual leave.

- o Coordinated refusal of any number of persons to continue to work or accept employment.
- o Refusal to work overtime, where such work is necessary for maintenance of essential defence services.
- o Any other conduct which results in, or is likely to result in, disruption of work in essential defence services.

Prohibition on strikes, lock-outs, and lay-offs:

- o Government may prohibit strikes, lock-outs, and lay-offs in units engaged in essential defence services.
- o It may issue such an order if necessary in the interest of sovereignty and integrity of India, security of any state, public order, public, decency and morality.

Penalties:

- o Persons commencing or participating in illegal strikes - Up to one year imprisonment or Rs 10,000 fine, or both.
- o All offences punishable will be cognisable and non-bailable.
- o Cognisable offences are those that require an immediate arrest.

- **Statement 3 is correct:** It will amend the Industrial Disputes Act, 1947 to include essential defence services under public utility services.

The undertakings which supply the basic necessary services such as electricity, water, gas, power, transport etc. comes under the purview of the public utility services.

Source:

<https://prsindia.org/billtrack/the-essential-defence-services-bill-2021>

Q.25) Ans: a

Exp:

- In a major boost to Atmanirbhar Bharat and strengthening Indian Army, Defence Research & Development Organisation (DRDO) successfully flight tested indigenously developed low weight, fire & forget Man-Portable Anti-Tank Guided Missile (MPATGM)
- The missile (Man-Portable Anti-Tank Guided Missile) has already been successfully flight-tested for the maximum range. The missile is incorporated with state-of-the-art Miniaturized Infrared Imaging Seeker along with advanced avionics.
- The man-portable missile is launched using a tripod designed for a maximum range of 2.5 km with a launch weight of less than 15 Kg, Control Flight Tests have been successfully carried out and Guided Flight Tests (with IIR Seeker) are planned.
- This missile is for infantry and Parachute (Special Forces) of the Indian Army. It is 'Soft' launched from a canister using an Ejection Motor. It uses a state-of-the-art IIR seeker for homing on to the target.



Source:

<https://www.livemint.com/news/india/drdo-successfully-flight-tests-home-grown-man-portable-anti-tank-guided-missile-11626864322755.html>

Q.26) Ans: c

Exp:

- **Statement 1 is correct:** A high strength titanium alloy has been developed by India's Defence Research and Development Organisation (DRDO). This alloy has been developed for applications in aerospace structural forgings. The titanium alloys are unique due to their ductility, higher strength, fracture toughness and fatigue which make them beneficial for various aircraft structures. These alloys are already being used by many developed nations in recent times as a beneficial substitute for the relatively heavier traditional Ni-Cr-Mo structural steels to achieve weight savings.
- **Statement 2 is correct:** Due to their high tensile strength to density ratio, high corrosion resistance, and ability to withstand moderately high temperatures without creeping, titanium alloys are used in aircraft, armor plating, naval ships, spacecraft, and missiles. For these applications titanium alloyed with aluminium, vanadium, and other elements is used for a variety of components including critical structural parts, fire walls, landing gear, exhaust ducts (helicopters), and hydraulic systems. In fact, about two thirds of all titanium metal produced is used in aircraft engines and frames.

Source

: <https://static.pib.gov.in/WriteReadData/userfiles/image/PIC1J66L.jpg>

Q.27) Ans: b

Exp:

- **Statement 1 is correct:** The Drone Rules have established an online platform hosted by the Directorate General of Civil Aviation ("DGCA"), for the management of various drone-related activities in India. The aim of the Rules is to create a 'digital sky platform' ("the Platform") which is a business-friendly single-window online system, with minimum human interference, where most of the permissions will be self-generated
- **Statement 2 is incorrect:** Drone has been defined as an unmanned aircraft system⁵ ("UAS"), that can operate autonomously or can be operated remotely without a pilot on board. The maximum all-up-weight for a drone is 500 Kilograms. In case, the weight is more than 500 kilograms, the provisions of the Aircraft Rules, 1937 shall apply instead of the Drone Rules. Drones have been classified based upon the maximum all-up weight (not more than 500 kilograms⁷) including payload as under –
 - Nano drone: Less than or equal to 250 grams;
 - Micro drone: Greater than 250 grams and less than or equal to 2 kilograms;
 - Small drone: Greater than 2 kilograms and less than or equal to 25 kilograms;
 - Medium drone: Greater than 25 kilograms and less than or equal to 150 kilograms; and
 - Large drone: Greater than 150 kilograms.
- **Statement 3 is correct:** The rules categorizes three zones ie Red, Yellow and Green.

Yellow zone (Intermediate zone)

o The airspace of defined dimensions above the land areas or territorial waters of India within which UAS operations are restricted and shall require permission from the concerned air traffic control



authority.

o The airspace above 400 feet or 120 metres in the designated green zone and the airspace above 200 feet or 60 metres in the area located between the lateral distance of 8 kilometres and 12 kilometres from the perimeter of an operational airport.

Red Zone (No-Fly Zone)

o The airspace of defined dimensions, above the land areas or territorial waters of India, or any installation or notified port limits specified by the Central Government beyond the territorial waters of India, within which UAS operations shall be permitted only by the Central Government.

o The Rules prohibit the operation of a drone in a red zone or yellow zone without prior permission. However, no such provision is required in the Green Zone.

Source: <https://pib.gov.in/PressReleaseDetailm.aspx?PRID=1749154>

Q.28) Ans: a

Exp:

24 MH-60 Romeo multirole helicopters

- Recently, The Indian Navy received three of 24 MH-60 Romeo multirole helicopters from the United States (US). India and the US had signed a deal worth \$2.4 billion in February 2020 to procure 24 MH-60 Romeo helicopters.
- MH-60R is the most advanced maritime multi-mission helicopter in operation - deployed globally, and its mission performance is, by far, second to none.
- The induction of these all-weather MRH would enhance the Indian Navy's three-dimensional capabilities. The helicopters would also be modified with several unique equipment and weapons.

Source: <https://www.indiatoday.in/india/story/indian-navy-receives-first-two-mh-60r-helicopters-from-us->

[1829217-2021-07-17](#)

Q.29) Ans: c

Exp:

- **Statement 1 is correct:** This single crystal blade technology has been developed by the Defence Metallurgical Research Laboratory (DMRL), one of the DRDO's laboratories. Few countries such as the US, the UK, France and Russia have the capability to design and manufacture single crystal components. The helicopters need powerful and compact aero engines to operate in extreme conditions. In order to achieve this, Single Crystal Blades are essential.
- **Statement 2 is correct:** Single Crystal Blade Technology is a "Grain Boundary" technology used in making blades. Here the metal forms grains while solidifying. Usually, metal alloys are used in making blades. Different metals in the alloy produce different grains while solidifying thereby creating grain boundaries. Basically, a grain boundary is an interface between two grains or crystals. These boundaries lead to creep in the blades.

- o The problem of creep is overcome in Single Crystal Blade technology by designing alloys that can be used in single crystal form. This means that no boundaries will be formed here.
- o The technology uses a vacuum investment casting process to realise the blades. This includes ceramic moulding, wax patterning, and heat treatment. It is essential to formulate special ceramic composition for the technology. This involves making strong ceramic moulds that withstands metallosstatic pressure at 1500 degrees Celsius.

Source

: <https://www.thehindu.com/news/national/drdo-develops-critical-crystal-blade-technology-for-aero->



[engines/article34415540.ece](https://www.indiatoday.in/world/story/india-russian-s-400-missiles-first-batch-october-december-rosoboronexport-ceo-1805066-2021-05-21)

Q.30) Ans: a

Exp:

- **Statement 1 is correct:** India's first indigenous anti-radiation missile, Rudram, developed for the Indian Air Force, was successfully flight-tested from a Sukhoi-30 MKI jet off the east coast. Anti-radiation missiles are designed to detect, track and neutralise the adversary's radar, communication assets and other radio frequency sources, which are generally part of their air defence systems.
- **Statement 2 is incorrect:** Rudram is an air-to-surface missile, designed and developed by the Defence Research and Development Organisation (DRDO). DRDO initiated development of anti-radiation missiles of this type around eight years ago, and its integration with fighter jets has been a collaborative effort of various DRDO facilities and formations of the IAF and Hindustan Aeronautics Ltd. While the system has been tested from a Sukhoi-30 MKI, it can be adapted for launch from other fighter jets too

Source

: <https://indianexpress.com/article/explained/why-anti-radiation-missile-rudram-matters-6718894/>

Q.31) Ans: c

Exp:

- The S-400 is known as Russia's most advanced long-range surface-to-air missile defence system. The 'Triumf' interceptor-based missile system can

destroy incoming hostile aircraft, missiles and even drones at ranges of up to 400 km.

- The radar can detect and track aircraft, rotorcraft, cruise missiles, guided missiles, drones and ballistic rockets within a distance of 600km. It can simultaneously track up to 300 targets.

<https://www.indiatoday.in/world/story/india-russian-s-400-missiles-first-batch-october-december-rosoboronexport-ceo-1805066-2021-05-21>

Q.32) Ans: a

Exp:

- **Statement 1 is correct:** The Defence Acquisition Council is the highest decision-making body in the Defence Ministry for deciding on new policies and capital acquisitions for the three services (Army, Navy and Air Force) and the Indian Coast Guard. It was formed, after the Group of Ministers recommendations on 'Reforming the National Security System', in 2001, post Kargil War (1999).
- **Statement 2 is incorrect:** A Defence Acquisitions Council has been created as an overarching structure with the following

Composition :-

- o Chairman: Raksha Mantri
- o Members: Raksha Rajya Mantris
- o Chief of Defence Staff (CDS) (when appointed) Chief of Army Staff
- o Chief of Naval Staff
- o Chief of Air Staff
- o Defence Secretary
- o Secretary Dept. of Defence Production and Supplies
- o Secretary, Defence Research and Development
- o Secretary, Defence Finance
- o Vice Chief of Defence Staff (when appointed)/CISC
- o Special Secretary (Acquisition)

The DAC will perform the following functions:



- Give 'in principle' approval to Capital acquisitions in the Long Term Perspective Plan (LTPP) covering a 15-year time span at the beginning of a Five Year Plan period.

- Give 'in principle' acceptance of necessity to each Capital acquisition project for incorporation in the forthcoming Five Year Plan, at least nine months before the commencement of the first year of that plan.

Source: https://www.mod.gov.in/dod/sites/default/files/DEFENCE_PROCUREMENT_ORGANISATION.pdf

Q.33) Ans: d

Exp:

Graphene

- Graphene is considered to be the world's thinnest, strongest and most conductive material - of both electricity and heat. The thermal and electrical conductivity is very high and it can be used as a flexible conductor.
- The electrical conductivity for doped graphene is potentially quite high. Graphene is practically transparent. It is thin, mechanically very strong, transparent and flexible conductor graphene has the potential to revolutionize entire industries - in the fields of electricity, conductivity, energy generation, batteries, sensors and more.
- Such graphene-enhanced composite materials can find uses in aerospace, building materials, mobile devices, and many other applications. This could be useful in both microelectronics (for example to make LED lighting more efficient and longer lasting) and also in larger applications - for example thermal foils for mobile devices
- Graphene is a very promising material for use in batteries and supercapacitors. Graphene may enable batteries and supercapacitors (and even fuel-cells) that can store more energy - and charge faster.

Source

<https://www.thehindu.com/sci-tech/Graphene-a-novel-material-with-myriad-uses/article15771016.ece/amp/>

Q.34) Ans: d

Exp:

Nanopollution

- Nanopollution is a generic name for all waste generated by nanodevices or during the nanomaterials manufacturing process. Nanowaste is mainly the group of particles that are released into the environment
- The basic theme of nanotechnology is to use particles having size in nanometer range for various applications in medical fields, cosmetics industry, and agriculture and food technologies.
- It is considered that NPs may pose risks to the environment and biological systems. It is also becoming evident that the size, structure and type of nanomaterials, such as graphene/graphene oxide with gold NPs, carbon and carbon nitride nanotubes, have different effects on plants and the environment.
- A recent study showed a high toxicity of carbon nanotubes which seemed to produce harmful effects by an entirely new mechanism, different from the normal model of toxic dusts.
- The pulmonary injury and inflammation resulting from the inhalation of nanosize urban particulate matter appears to be due to the oxidative stress that these particles cause in the cells. Persistent insoluble nanoparticles may cause problems in the environment that are much greater than those revealed by human health assessments.

Source

<https://indianexpress.com/article/technology/science/i>



[ndian-scientists-work-on-nano-material-paves-way-for-clean-energy-vehicle-6201426/lite/](https://www.pib.gov.in/Press-Release-Details.aspx?PRID=1426)

Q.35) Ans: c

Exp:

- **Both 1 and 2 statements are correct** : The Indian government aims to grow its nuclear power capacity as part of its infrastructure development programme, largely indigenous nuclear power programmes. **Nuclear power is the fifth-largest source of electricity** in the country after coal, gas, hydroelectricity and wind power.
- The **Atomic Energy Act of 1948 created the Indian Atomic Energy Commission** "to provide for the development and control of atomic energy and purposes connected therewith". India now has 23 operable nuclear reactors and seven nuclear reactors under construction.
- **Because of its weapons programme, India is outside the Nuclear Non-Proliferation Treaty.** Due to this, the country was mostly excluded from nuclear plants and material trade.

Source - <https://www.indiatoday.in/amp/education-today/gk-current-affairs/story/facts-on-india-s-nuclear-power-and-list-of-nuclear-research-centres-in-india-1866649-2021-10-19>

Q.36) Ans: b

Exp:

- **Statement 1 and 2 are correct** : GICNT is an international partnership intended to improve international capacity for prevention, detection and response to nuclear terrorism, particularly the acquisition, transportation or utilization of nuclear and radiological materials.
- Russian Federation President Vladimir Putin and United States President George W. Bush jointly

announced the **creation of the Global Initiative to Combat Nuclear Terrorism (GICNT) during the G8 Summit in St. Petersburg, Russia on 15 July 2006.** The Initiative builds on the Convention for the Suppression of Acts of Nuclear Terrorism, the Convention on the Physical Protection of Nuclear Material and its 2005 Amendment, and United Nations Security Council Resolutions 1373 and 1540.

- If a **State desires to become a partner in the GICNT it must send a written letter of endorsement of the Statement of Principles to the GICNT.**
- **Statement 3 is incorrect** : Currently, the international Atomic Energy Agency (IAEA), the European Union (EU), INTERPOL, UNOCT, and the United Nations Office on Drugs and Crime (UNODC) are official observers of the GICNT. Only international governmental organizations can serve as official observers of the GICNT.

source - <https://www.nti.org/education-center/treaties-and-regimes/global-initiative-combat-nuclear-terrorism-gicnt/>

Q.37) Ans: d

Exp:

Nuclear energy

- Nuclear energy generates about 10 percent of the world's electricity, and after hydropower, it is the world's second largest source of low-carbon power, according to the International Energy Agency (IEA). Nuclear energy can also be utilized to decarbonize non-electric applications. The heat produced by nuclear power plants is used to create steam, which drives electricity-generating turbines. Existing nuclear fleets today reach operating temperatures in the range of 300°C, while district heating and seawater desalination processes require about 150°C.



- Further, Heat produced by nuclear power plants can be utilized for heating or cooling, or as an energy source towards the production of fresh water, hydrogen or other products, such as oil or synthetic fuel.
- There are several methods to use nuclear energy, as a source of electricity and heat, to produce hydrogen efficiently and with little to no CO₂ emissions.
- In nuclear district heating, steam produced by a nuclear power plant serves to heat regional heating networks.
- The Heating Project is an example of how nuclear energy can play a role in decarbonizing residential heating, as well as the added value of operating a nuclear power plant in cogeneration mode.

Source - <https://www.iaea.org/newscenter/news/the-use-of-nuclear-power-beyond-generating-electricity-non-electric-applications>

Q.38) Ans: a

Exp:

- **Statement 1 is incorrect** : "The Large Hadron Collider (LHC)" is the world's largest and highest-energy particle collider. It was built by the European Organization for Nuclear Research (CERN) between 1998 and 2008 in collaboration with over 10,000 scientists and hundreds of universities and laboratories, as well as more than 100 countries. It lies in a tunnel 27 kilometres (17 mi) in circumference and as deep as 175 metres (574 ft) beneath the France–Switzerland border near Geneva.
- **Statement 2 is correct** : The LHC's goal is to allow physicists to test the predictions of different theories of particle physics, including measuring the

properties of the Higgs boson searching for the large family of new particles predicted by supersymmetric theories and other unresolved questions in particle physics. The Standard Model of particle physics predicts that about 60% of the time a Higgs boson will decay to a pair of bottom quarks, the second-heaviest of the six flavours of quarks.

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"Higgs boson" is an elementary particle in the Standard Model of particle physics, **produced by the quantum excitation of the Higgs field**, one of the fields in particle physics theory.

It is named after **physicist Peter Higgs, who in 1964, along with six other scientists**, proposed the mechanism, which suggested the existence of such a particle.

Its existence was **confirmed by the ATLAS and CMS collaborations based on collisions in the LHC at CERN.**

Source - <https://www.thehindu.com/sci-tech/science/lhc-new-physics-in-sight/article7411360.ece/amp/>

Q.39) Ans: c

Exp:

- **Statement 1 is incorrect**: Protons, neutrons, and electrons are tiny particles that make up atoms. The **neutrino is also a tiny elementary particle, but it is not part of the atom. Neutrinos have a very tiny mass, no charge and spin half.**
- **Statements 2,3,4 are correct** : Neutrinos come from the sun (solar neutrinos) and other stars, cosmic rays that come from beyond the solar system, and from the Big Bang from which our Universe originated. **They can also**



be produced in the lab. The India-based Neutrino Observatory (INO) will study atmospheric neutrinos only. Solar neutrinos have much lower energy than the detector can detect. Atmospheric neutrinos are produced from cosmic rays which consist of protons and heavy nuclei. These collide with atmospheric molecules such as Nitrogen to give off pions and muons which further decay to produce neutrinos.

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Neutrinos come in **three** types or “flavours” – electron neutrino, tau neutrino and muon neutrino.

They can change from one flavor to another as they travel. This process is called neutrino oscillation and is an unusual quantum phenomenon.

Following can be possible future **applications** of “neutrino science”.

Information about properties of the sun

In astronomy to discover the composition of the universe.

Medical Imaging

A way to monitor nuclear proliferation

Faster global communication

Source - <https://www.thehindu.com/sci-tech/science/what-are-neutrinos-and-how-are-they-detected/article23546887.ece/amp/>

Q.40) Ans: d

Exp:

- **All Statements are correct**

- The visible light that reaches us from the sun is emitted from the surface of the sun. The neutrinos which also take close to this time to reach us from the sun, known as “solar neutrinos,” were produced in the core of the sun. They give us information about the interior of the sun. Studying these neutrinos can help us understand what goes on in the interior of the sun.

- Light coming from distant stars can be studied by astronomers, for example, to detect new planets. Light is the visible part of the electromagnetic spectrum, other parts are used in, for example, radio astronomy. Likewise, if the properties of neutrinos are understood better, they can be used in astronomy to discover what the universe is made up of.
- The extragalactic neutrinos we observe may be coming from the distant past. These inviolate messengers can give us a clue about the origin of the universe and the early stages of the infant universe, soon after the Big Bang.
- They can be used in technological applications of the detectors that will be used to study X-ray machines, PET scans, MRI scans, etc., all of which came out of research into particle detectors.

Source - <https://www.thehindu.com/sci-tech/science/what-are-neutrinos-and-how-are-they-detected/article23546887.ece/amp/>

Q.41) Ans: c

Exp:

- **Statement 2 and 4 are Incorrect :** The Nuclear Command Authority comprises a Political Council and an Executive Council. The Political Council is chaired by the Prime Minister. It is the sole body which can authorise the use of nuclear weapons. The Executive Council is chaired by the National Security Advisor. It Provides inputs for decision making by the Nuclear Command Authority and executes the directives given to it by the Political Council. **Hence Statement 1 and 3 are correct.**
 - India adopted its nuclear doctrine in 2003. The salient features of this doctrine are as follows:



- Building and maintaining a credible minimum deterrence.
- A posture of “**No First Use**” – nuclear weapons will only be used in retaliation against a nuclear attack on Indian territory or on Indian forces anywhere.
- Nuclear retaliation to a first strike will be massive and designed to inflict unacceptable damage.
- Nuclear retaliatory attacks can only be authorised by the civilian political leadership through the Nuclear Command Authority.
- Non-use of nuclear weapons against non-nuclear weapon states.
- However, in the event of a major attack against India, or Indian forces anywhere, by biological or chemical weapons, India will retain the option of retaliating with nuclear weapons.
- A continuance of strict controls on export of nuclear and missile related

materials and technologies, participation in the Fissile Material Cutoff Treaty negotiations, and continued observance of the moratorium on nuclear tests. Continued commitment to the goal of a nuclear-weapon-free world, through global, verifiable and non-discriminatory nuclear disarmament.

Q.42) Ans: c

Exp:

Flux trapping Effect

- "Superconductivity" which relates to the electrical resistance a material exhibits. In a superconductor, or a material with superconductivity, this resistance is zero ohms or very near zero. Generally, this involves the cooling of a material to ultra low temperatures, such as below 20°K (-253°C). The cooling temperature varies between materials and is known as the transition temperature.
- When a magnet is placed directly over a superconductor it will display properties of levitation. If the magnet is removed, the

superconducting material will still display these properties. This is known as the flux trapping effect.

- The superconductivity phenomenon was discovered in 1911 by Dutch physicist Heike Kamerlingh Onnes. Like ferromagnetism and atomic spectral lines, superconductivity is a phenomenon which can only be explained by quantum mechanics.
- This can be achieved at very low temperatures and high pressure .

Extra Edge by Only IAS

- Some of the technological applications of superconductivity include:
- the production of sensitive magnetometers based on SQUIDs (superconducting quantum interference devices)
- fast digital circuits (including those based on Josephson junctions and rapid single flux quantum technology)
- powerful superconducting electromagnets used in maglev trains, magnetic resonance imaging (MRI) and nuclear magnetic resonance (NMR) machines, magnetic confinement fusion reactors (e.g. tokamaks), and the beam-steering and focusing magnets used in particle accelerators
- low-loss power cables.

- RF and microwave filters (e.g., for mobile phone base stations, as well as military ultra-sensitive/selective receivers)

Source - <https://www.scienceforums.net/topic/26213-flux-trapping-effect-of-superconductors/>

Q.43) Ans: c

Exp:



- **The National Physical Laboratory of India**, situated in New Delhi, is the measurement standards laboratory of India. It maintains standards of SI units in India and calibrates the national standards of weights and measures. It was established on 4 January 1947, India's first national laboratory.
- **The Atomic Energy Commission of India** is the governing body of the Department of Atomic Energy (DAE), Government of India. It was formed on 3 August 1948. The DAE is under the direct charge of the Prime Minister. A resolution passed by the Government of India later replaced the commission by "Atomic Energy Commission of India" on 1 March 1958 under the Department of Atomic Energy with more financial and executive powers. Its headquarters is in Mumbai, Maharashtra. Functions of the Atomic Energy Commission are as follows :
 - To organize research in atomic science in the country
 - To train atomic scientists in the country
 - To promote nuclear research in the commission's own laboratories in India
 - To undertake prospecting of atomic minerals in India and to extract such minerals for use on an industrial scale.
 - Indian Institute of Technology Kharagpur (IIT Kharagpur) is a public technical and research university established by the government of India in Kharagpur, West Bengal, India. Established in 1951, the institute is the first of the IITs to be established and is recognised as an Institute of National Importance. In 2019 it was awarded the status of Institute of Eminence by the government of India. The institute was initially established to train scientists and engineers after India attained independence in 1947.
- **DRDO**- The Defence Research and Development Organisation. It is the premier agency under the

Department of Defence Research and Development in the Ministry of Defence of the Government of India, charged with the military's research and development, headquartered in Delhi, India. It was formed in 1958 by the merger of the Technical Development Establishment and the Directorate of Technical Development and Production of the Indian Ordnance Factories with the Defence Science Organisation. **DRDO is India's largest and most diverse research organisation.**

Q.44) Ans: d

Exp:

- Recently, Apollo Proton Cancer Centre (APCC), India has become the 16th country in the world to offer proton therapy for cancer.
- The centre offers proton therapy with pencil-beam technology that provides a high degree of precision.
- Proton therapy can be utilised for young adults and teenagers to reduce the risk of second cancer.
- It uses protons rather than x-rays to treat cancer. At high energy, protons can destroy cancer cells.
- It can also be combined with x-ray radiation therapy, surgery, chemotherapy, and/or immunotherapy.
- Like x-ray radiation, proton therapy is a type of external-beam radiation therapy.
- It is the most technologically advanced method to deliver radiation treatments to cancerous tumors.



Extra Edge by Only IAS

Proton therapy also may be used to treat these cancers:

Central nervous system cancers, including chordoma, chondrosarcoma, and malignant meningioma

Eye cancer, including uveal melanoma or choroidal melanoma

Head and neck cancers, including nasal cavity and paranasal sinus cancer and some nasopharyngeal cancers

Lung cancer

Liver cancer

Prostate cancer

Spinal and pelvic sarcomas, which are cancers that occur in the soft-tissue and bone

Non Cancerous brain tumors

Source

<https://www.thehindu.com/news/cities/chennai/apollo-hospitals-inaugurates-proton-cancer-centre/article26083995.ece/amp/>

Q.45) Ans: d

Exp:

Applications of Superconductivity

- Superconducting magnets are used for accelerating the particles in the Large Hadron Collider.
- SQUIDS (superconducting quantum interference devices) are being used in the production of highly sensitive magnetometers. They are generally used for the detection of landmines.
- Superconducting magnets are also used in Magnetic Resonance Imaging (MRI) machines.

- As we know due to the electrical resistance, there is a power loss while power transmission. So nowadays, superconducting cables are used in place of ordinary cable lines to avoid power loss.
- Superconductors are also being used for the development of high-intensity Electromagnetic Impulse (EMP).
- Maglev trains work on the superconducting magnetic levitation phenomenon. The Japanese Maglev train is a real-life example of magnetic levitation.

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The superconductivity phenomenon was discovered in 1911 by Dutch physicist Heike Kamerlingh Onnes. Like ferromagnetism and atomic spectral lines, superconductivity is a phenomenon which can only be explained by quantum mechanics.

This can be achieved at **very low temperatures and high pressure**.

Source

<https://physicsinmyview.com/2019/06/superconductor-definition-types-uses.html>

Q.46) Ans: d

Exp:

- The **technical cooperation programme is the IAEA's primary mechanism** for transferring nuclear technology to Member States, helping them to address key development priorities in areas such as health and nutrition, food and agriculture, water and the environment, industrial applications, and nuclear knowledge development and management.



- The programme also helps Member States to identify and meet future energy needs, and assists in improving radiation safety and nuclear security worldwide, including through the provision of legislative assistance.

Extra Edge by Only IAS

International Atomic Energy Agency (IAEA) The IAEA is the international centre for cooperation in the nuclear/atomic field. It is a **UN agency**. It works with its member countries and many partners to promote peaceful uses of nuclear technologies.

It is also known as '**Atoms for Peace**' and this was the organisation's first name when it was formally **established in 1957**.

Headquartered in Vienna, Austria, the IAEA is a UN agency

Source

<https://www.iaea.org>

Q.47) Ans: a

Exp:

- **Statement 1 is correct** : Nuclear Suppliers Group is a **transnational body** composed of nuclear supplier countries that aim to **control the proliferation of nuclear weapons** by curbing the export of nuclear weapons development materials and related technology. It seeks to **improve the existing safeguards on existing nuclear materials**. The group had its first meeting in November 1975. A series of meetings held in London produced agreements on export guidelines. Although **initially there were only 7 countries as members**.
- **Statement 2 is incorrect** : India is not a member of the Nuclear Suppliers Group. The NSG guidelines require

that *importing states provide assurances to NSG members that proposed deals will not contribute to the creation of nuclear weapons*. **Certain eligibility needs to be fulfilled for a country to become a member of the NSG.** The use of nuclear fuel available for peaceful nuclear activities that are subject to **International Atomic Energy Agency (IAEA) safeguards**, as well as for other industrial activities where they would not contribute to nuclear proliferation.

Source

- <https://www.thehindu.com/cdn.ampproject.org/v/s/www.thehindu.com/news/national/indias-entry-into-elite-nuclear-groups->

Q.48) Ans: d

Exp:

The **NSG controls the global exports of nuclear technology and material to ensure that atomic energy is used only for peaceful purposes**.

- Given below are the reasons how joining NSG will be beneficial for India:
- It will provide the country with access to foreign-sourced nuclear material and equipment, reducing the risk faced by foreign nuclear industries in doing business with India.
- Increasing the business of these nuclear materials will enable India to make better versions of nuclear breeders and export them to smaller countries, thereby increasing the economic growth of the country.
- The Make In India program will also see a boost if India becomes a member of NSG because nuclear power production would increase.
- It will also give India an opportunity to initiate talks about the plutonium trade for its thorium program and gain massive domestic profits.
- India aims to minimize the use of fossil fuels by 40 percent and use more natural and renewable



resources of energy. This is possible if India gets access to nuclear raw materials and increases nuclear power generation.

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India becoming a member of the 'Australia Group' (AG), a move that is expected to raise New Delhi's stature in the field of non-proliferation and also help it acquire critical technologies .India is now a member of the Missile Technology Control Regime (MTCR), the Wassenaar Arrangement (WA) as well as the AG, three of four non-proliferation regimes. The only one remaining is the Nuclear Suppliers Group (NSG).

Source

<https://www.thehindu.com.cdn.ampproject.org/v/s/www.thehindu.com/news>

Q.49) Ans: a

Exp:

- Statement 1 correct The NPT is an international treaty whose objective is to prevent the spread of nuclear weapons and weapons technology, to foster the peaceful uses of nuclear energy, and to further the goal of disarmament. The treaty was signed in 1968 and entered into force in 1970. Presently, it has 190 member states.
- Nuclear-weapon states parties under the NPT are defined as those that manufactured and exploded a nuclear weapon or other nuclear explosive device before January 1, 1967.
- Despite being a non-party, India abides by the principles and objectives of the NPT, including its nuclear disarmament aspirations, he said, adding that India is committed to making its contribution to strengthening non-proliferation.

- Statement 2 is incorrect: With its near-universal membership, the NPT has the widest adherence of any arms control agreement, with only South Sudan, India, Israel, and Pakistan remaining outside the treaty. The treaty, which was indefinitely extended in 1995, calls for a review conference every five years to assess progress on achieving the treaty's key objectives and provide opportunities to discuss new measures to strengthen the treaty.

Source

<https://www.thehindu.com/news/national/india-rules-out-joining-npt-as-non-nuclear-weapon-state/article19855611.ece/amp/>

Q.50) Ans: a

Exp:

- India's first Lithium refinery which will process Lithium ore to produce battery-grade material will be set up in Gujarat.
- With India poised to become one of the largest electric car markets of the world, the country is also looking to access raw materials like Lithium needed to produce batteries. India currently imports most of the Lithium needed. China, Hong Kong and Vietnam were the leading sources of imports.

Extra edge by only IAS

Lithium:

Lithium is a soft, white, lustrous and alkali metal which is found in brine deposits and salts.

Largest producer of lithium in the world- **Australia**
Lithium Triangle Countries: it is located in South America. The three countries that make up the Lithium Triangle are- **Bolivia, Argentina and Chile.**



Q.51) Ans: c

Exp:

Option C is correct:

- Offshore wind energy refers to the deployment of wind farms inside the water bodies. They utilise the sea winds to generate electricity.
- Onshore wind energy refers to the deployment of wind farms on land.
- Benefit of Offshore wind energy:
- Absence of any obstruction in the sea.
- Wind speed is high and is consistent in direction.
- Offshore wind farms have a higher 'capacity utilisation factor' CUF than onshore wind farms.
- In India, where land is limited and the population is increasing, large wind farms positioned over water bodies will be vital.
- Maintenance and installation of Offshore wind turbines is more costly than onshore wind turbines.

Source- downtoearth.com

Q.52) Ans: d

Exp:

- Solar energy is the radiation from the Sun capable of producing heat, causing chemical reactions, or generating electricity.
- Solar energy technology is the technology which is used to convert solar radiation into electricity.
- Positive implication of the use of solar energy Technologies:
- - Electricity generation.
 - Production of thermal energy (heating or cooling, either through passive or active means).
 - Meeting direct lighting needs.
 - Produce fuels that might be used for transport

and other purposes.

- Reduction of the emission of the greenhouse gasses (mainly CO₂, NO_x) and of toxic gas emissions (SO₂, particulates)
- Reclamation of degraded land.
- Reduced requirement for transmission lines within the electricity grid.
- Improvement of the water resources quality.

Source- sciencedirect.com

Q.53) Ans: c

Exp:

Both the statements are correct:

- Solar energy technologies harness the energy of solar irradiance to produce electricity.
- Currently, there are principally two technologies employed: photovoltaics (PV) and concentrating solar power (CSP) technologies.
- **Photovoltaics Technology** (PV) generates electricity using the conducting properties of certain chemicals, most importantly **silicon**, through the photo-electric effect.
- **Concentrating solar power Technology** (CSP) makes use of reflectors to focus sunlight on a small area, in order to generate steam that powers a thermal electric plant.

Source- sciencedirect.com

Q.54) Ans: a

Exp:

Silicon semiconductors

- Conversion of solar energy into electric energy is accomplished by the principle of photovoltaic (PV) effect.



- One of the most popular semiconductor materials used in PV cells is silicon which has properties conducive to produce electric power from solar energy.

Silicon semiconductors are popular because:

- Low weight volume ratio.
- Extended life cycle.
- Robustness and strength.
- Silicon semiconductors are easy to produce at low cost.
- High quality semiconductor which has properties conducive to produce electric power from solar energy.
- Silicon makes up 27.7 percent of Earth's crust; it is the second most abundant element in the crust, being surpassed only by oxygen.

Source- economictimes.com

Q.55) Ans: a

Exp:

Option A is correct:

- **Stratospheric aerosol injection (SAI):** Gases are pumped into the stratosphere to reflect some of the sun's heat, mimicking an effect that happens naturally in a strong volcanic eruption.
- **Marine cloud brightening (MCB):** MCB involves reflecting sunlight away from the earth in some way. In this case, sea salt or other particles are sprayed into marine clouds to make them thicker and more reflective.
- **Ground-based albedo modification (GBAM):** The word albedo refers to how much solar radiation a planet reflects from its surface. Modifying earth's albedo from the ground could come from small

architectural measures, like whitening roofs and land-use management. It could also take on a much larger scale; ideas mentioned in the IPCC report include covering glaciers or deserts with reflective sheeting.

Source- ipcc.ch

Q.56) Ans: a

Exp:

- **Carbon Dioxide Removal (CDR)** as a climate-change mitigation measure that is incorporated into nearly all of its safe pathways forward.
- CDR tackles the root cause of global warming by removing greenhouse gases from the atmosphere — like 'carbon capture and storage'.
- CDR refers to different techniques for removing carbon dioxide (CO₂) from the atmosphere, reducing warming. The most established of these processes is bioenergy with carbon capture and storage (BECCS).
- BECCS involves burning biomass for energy, and capturing and storing the emissions underground.
- Alternative forms of CDR include direct air carbon dioxide capture and storage (DACCS). This was used in a commercial operation by ClimeWorks that opened last year.
- **Soil carbon sequestration** (SCS), afforestation and ocean fertilisation, where added nutrients cause CO₂ to be trapped in the deep ocean, are also methods of CDR. Deforestation is not the carbon dioxide removal method from atmosphere, even deforestation increased the the carbon dioxide emissions indirectly in the atmosphere by reduce the carbon sink.
- The chemical capture and storage of CO₂ deep underground is seen as generally secure for thousands of years. Natural carbon sinks like soil



and trees reverse earlier, so they need to be used with care.

Source- dezeen.com

Q.57) Ans: a

Exp:

- **Option A is correct:** As a part of Nationally Determined Contributions as per the Paris Accord on Climate Change, 2015 India has made a pledge that by the year 2022, 175 GW of renewable energy capacity will be installed. This includes 100 GW from solar, 60 GW from wind, 10 GW from biomass and 5 GW from small hydro power.

India's Renewable Energy Target by 2022 (By Source)

Unit: Gigawatts



- The substantial higher capacity target will ensure greater energy security, improved energy access and enhanced employment opportunities

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Renewable energy installation target of India:

- By 2022: Total 175GW renewable energy to be installed.
- By 2030: Total 450GW renewable energy to be installed.

Rank of India in the world:

- 4th Position: In terms of installed Total Renewable Energy capacity.
- 5th Position: In terms of installed Solar Energy capacity.
- 4th Position: In terms of installed Wind Energy capacity.

Intended nationally determined contributions (INDC):

- INDC is the set of targets of countries to reduce greenhouse gas emissions under the UNFCCC.
- INDC was submitted during the COP-21 of the UNFCCC, in which the Paris Agreement was also signed.
- INDC India submitted a target of 33-35% reduction of greenhouse gas per unit of GDP by 2030 from 2005 level.
- To achieve about 40% cumulative electric power installed capacity from non fossil fuel based energy resources by 2030, with the help of transfer of Technology and low cost International finance, including from Green Climate Fund.
- To create an additional carbon sink of 2.5 to 3 billion tonnes of CO₂ equivalent through additional forest and tree cover by 2030.

Q.58) Ans: b

Exp:

- **Statement 1 is Incorrect:** Membership of the International Solar Organization preferably available only for solar resource rich countries (which lie either completely or partly between the Tropic of Cancer and the Tropic of Capricorn) to address their special energy needs. But after the amendment in 2021 those countries lying beyond tropics can join the ISA.



- **Statement 2 is correct:** By the amendment in 2021 all member states of the United Nations including those lying beyond the tropics can now join the international solar Alliance.

Extra edge by only IAS **International solar Alliance:**

The International Solar Alliance was launched jointly by the Indian Prime Minister and the French President during the 2015 United Nations Climate Change Conference, or COP21, in Paris.

It aims to contribute to the implementation of the Paris Climate Agreement through rapid and massive deployment of solar energy.

The ISA aims to bring together countries to provide a collective response to the main common obstacles to the massive deployment of solar energy in terms of technology, finance and capacity.

The objective of the ISA is to mobilise member countries, seek commitments from international organisations and mobilise the private sector, to support rural and decentralised applications, access to affordable finance, island and village solar mini grids, rooftop installations, and solar e-mobility technologies.

The international solar Alliance is the **first international body to have a secretariat in India** although it was founded in Paris France its headquarter is in **gurugram, Haryana India**.

Q.59) Ans: d

Exp:

- **Statement 1 is correct:** Quantum computers store and process information using quantum two level systems (quantum bits or qubits) which unlike classical bits, can be prepared in superposition states. This key ability makes quantum computers extremely powerful compared to conventional

computers when solving certain kinds of problems like finding prime factors of large numbers and searching large databases.

- **Statement 2 is correct:** A Quantum computer uses Qubits which is described as a 0 and 1 at the same time giving us more processing power compared to a classical computer. In a classical computer information is expressed through a single number either 0 or 1.
- **Statement 3 is correct:** A Quantum computer uses the laws of quantum mechanics in which superposition and entanglement are what give quantum computers the ability to process so much more information.

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Superposition: It means that the **Qubit (Quantum bit)** which is the basic unit of information in a Quantum computer, can represent **both 0 and 1 at the same time** and give us more processing power.

Entanglement: It means **two subatomic particles** at the opposite end of the universe are **entangled** (linked/tied) in such a way that **any changes in one disturbs the other particle automatically**.

Q.60) Ans: a

Exp:

Option A is correct:

- Recently, **Google's quantum computer, named Sycamore**, claimed '**Quantum Supremacy**' because it reportedly did the task in 200 seconds that would have apparently taken a supercomputer 10,000 years to complete.
- Quantum supremacy refers to quantum computers being able to solve a problem that a classical

computer cannot. And that problem can be solved by a supercomputer in 10,000 years approx.

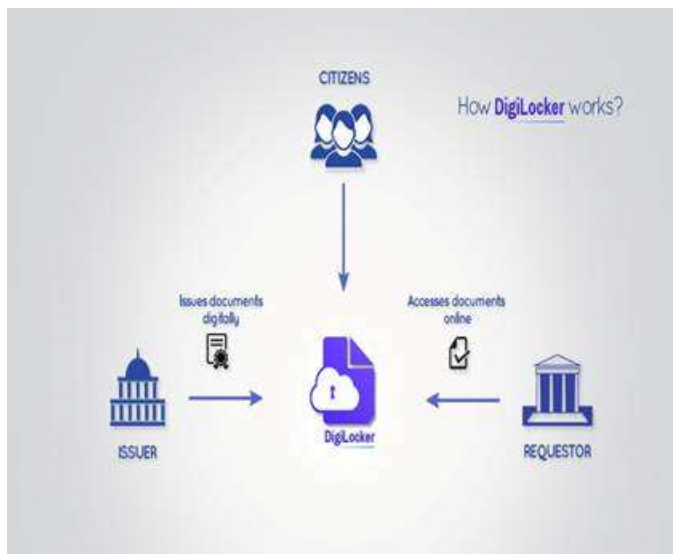
Q.61) Ans: d

Exp:

About DigiLocker

- DigiLocker (digital locker) is a flagship initiative under the digital India mission which was launched by the Ministry of Electronics and Information Technology Government of India, in 2015.
- Applications of DigiLocker:

• DigiLocker is a platform for issuance and verification of documents and certificates in a digital way.



• Organizations that are registered with Digital Locker can push electronic copies of documents and certificates directly into citizens' lockers. **For example-** This year CBSE uploaded the 10th result directly into digilocker.

• Citizens can also upload scanned copies of their legacy documents in their accounts.

• Legacy documents can be electronically signed using the eSign facility.

• DigiLocker account users get a dedicated cloud storage space that is linked to their Aadhaar (UIDAI) number.

• **Benefit of digital locker:**

- Citizens can access their digital documents anytime, anywhere.
- Citizens can also share their documents online.
- It reduces the administrative overhead of Government departments by minimizing the use of paper.
- Digital Locker makes it easier to validate the authenticity of documents as they are issued directly by the registered issuers.
- Self-uploaded documents can be digitally signed using the eSign facility.

Source- <https://www.meity.gov.in/digital-locker>

Q.62) Ans: d

Exp:

- **Statement 1 is correct:** It is the high performance computing artificial intelligence (HPC-AI) supercomputer installed in the Centre for Development of Advanced Computing (C-DAC) unit. These super computers are used by academic researchers, MSME's and startups in areas like Oil exploration, flood prediction as also genomics and drug Discovery.
- **Statement 2 is correct:** it is established under the National Supercomputing Mission. The National Supercomputing Mission was launched to enhance the research capacities and capabilities in the country by connecting them to form a Supercomputing grid.
- **Statement 3 is correct:** India's newest and fastest supercomputer, PARAM-Siddhi AI, has been ranked 62nd in the TOP-500 list of most powerful supercomputers in the world. The TOP-500 project



tracks the most powerful supercomputers in the world, and is published twice a year.

Source-

<https://pib.gov.in/PressReleasePage.aspx?PRID=1709875>

Q.63) Ans: c

Exp:

National Supercomputing Mission

- The Mission would be implemented and steered jointly by the Department of Science and Technology (DST) and Department of Electronics and Information Technology (DeitY) and it is implemented by the Centre for Development of Advanced Computing (C-DAC), Pune, and the Indian Institute of Science (IISc), Bengaluru.

Objective of NSM

- To make India one of the world leaders in Supercomputing and to enhance India's capability in solving grand challenge problems of national and global relevance.
- To empower our scientists and researchers with state-of-the-art supercomputing facilities and enable them to carry out cutting-edge research in their respective domains.
- To minimize redundancies and duplication of efforts, and optimize investments in supercomputing.
- To attain global competitiveness and ensure self-reliance in the strategic area of supercomputing technology.

Source- <https://nsmindia.in/>

Q.64) Ans: d

Exp:

According to National supercomputing mission following are application areas of Supercomputers:

- Climate Modelling
- Weather Prediction
- Aerospace Engineering including CFD, CSM, CEM
- Computational Biology
- Molecular Dynamics
- Atomic Energy Simulations
- National Security/ Defence Applications
- Seismic Analysis
- Disaster Simulations and Management
- Computational Chemistry
- Computational Material Science and Nanomaterials
- Discoveries beyond Earth (Astrophysics)
- Large Complex Systems Simulations and Cyber Physical Systems
- Big Data Analytics
- Finance
- Information repositories/ Government Information Systems

Source- <https://nsmindia.in/>

Q.65) Ans: a

Mass Rapid Transportation Technologies

The ministry of road transport and highways sought the approval of Niti Aayog to experiment and introduce 06 latest mass rapid transportation technologies. These technologies include metrolino, stadler buses, hyperloop, pod taxis, hybrid buses and freight rail road.

Benefits of Mass Rapid Transportation Technologies:

- Resolve the increasing traffic crisis in the world.
- Cost effective than the existing ones.



- Something like metrino (pod taxis) is almost one tenth of the cost of the metro trains.
- Reduce air pollution and CO₂ emission.
- Encourage higher density development.
- Better use of expensive urban space.
- Promote greater equity and mobility.

Source- economictimes.com

Q.66) Ans: b

Exp:

Statement 1 is incorrect: In March 2021, the **Ministry of Civil Aviation** (MoCA) published the UAS(Drones) Rules, 2021. Drones can be significant creators of employment and economic growth due to their reach, versatility, and ease of use, especially in India's remote and inaccessible area.

Some important features of Drone Rules 2021:

- Digital sky platform shall be developed as a user-friendly single-window system. regime.
- Interactive airspace maps with green, yellow and red zones shall be displayed on the digital sky platform.
- No permission required for operating drones in green zones.
- No requirement for security clearance before issuance of any registration or licence.
- No remote pilot licence required for micro drones (for non-commercial use) and nano drones.
- No requirement of Type Certificate, unique identification number.
- No restriction on foreign ownership in Indian drone companies.
- Import of drones to be regulated by DGFT.
- Requirement of import clearance from DGCA abolished.

- Coverage of drones under Drone Rules, 2021 increased from 300 kg to 500 kg. This will cover drone taxis also.
- Nano and model drones (made for research or recreation purposes) are exempt from type certification.
- Maximum penalty for violations reduced to INR 1 lakh.
- Safety and security features like 'No permission – no takeoff' (NPNT), real-time tracking beacon, geo-fencing etc. to be notified in future.
- Drone corridors will be developed for cargo deliveries.
- Drone Promotion Council (DPC) to be set up by the Government with participation from academia, startups and other stakeholders to facilitate a growth-oriented regulatory regime.

Source- pib.gov.in

Q.67) Ans: c

Exp:

Option C is correct: The **Ministry of Civil Aviation** has launched an airspace map of India for drone operations:

S. N.	Operating Zone	Regulations/Permissions
1	Green Zone	In green zones, no permission whatsoever is required for operating drones with an all-up weight upto 500 kg.
2	Yellow Zone	Drone operations in yellow zone require permission from the concerned Air Traffic Control authority – AAI, IAF, Navy, HAL



		etc.as the case may be. NOT from the concerned State Government.
3	Red Zone	Red zone is the 'no-drone zone' within which drones can be operated only after a permission from the Central Government .

Source-

<https://pib.gov.in/PressReleaselframePage.aspx?PRID=1757850>

Q.68) Ans: d

Exp:

Unmanned Aircraft Systems (UAS), commonly known as drones, offer tremendous benefits to almost all sectors of the economy like:

- Agriculture- soil health assessment and resource utilisation.
- Mining- monitoring of mining areas and different encroachment.
- Infrastructure Development.
- Surveillance
- Emergency Response- Disaster management, Search and rescue, exact and fast delivery of essential goods.
- Transportation- traffic management.
- Geo-spatial mapping
- Defence Sector- real time surveillance, security planning.
- Effective Law Enforcement.

Source- pib.gov.in

Q.69) Ans: a

Exp:

- **Pegasus is a spyware** developed by NSO Group, an Israeli surveillance firm, that helps spies hack into phones. In July 2021, Amnesty International, along with 13 media outlets across the globe released a report on how the spyware was used to snoop hundreds of individuals, including Indians.

Q.70) Ans: d

Exp:

- **Virtual reality (VR)** - A fully immersive experience where the user leaves the Real World Environment behind to inter a fully digital environment via VR headset. In VR actual Or real world replaced by 3D digital elements. VR applications are now used in many sectors and industries like- military, engineering, construction, healthcare, education etc.
- **Augmented Reality (AR)** - An experience where virtual objects are superimposed/overlays onto their real-world environment. In Augmented Reality the actual/real world environment is not replaced with 3D digital elements. Most well known application of AR is the Pokemon GO game.
- **Mixed Reality (MR)** - Virtual objects placed in the real world can be interacted with and respond as if they were actual objects. It is the combination of virtual reality and Augmented reality.

Source- forbes.com

Q.71) Ans: a

Exp:



- Metaverse is the next generation of the internet. The metaverse is a form of mixed reality, meaning the combination of augmented and virtual reality. Metaverse will not only introduce digital elements in the real world, but it will also merge the Internet with the virtual world.
- It is the successor of the mobile internet where people can do more than what they do now on the Internet. People can dance and hang out with friends almost like in real time.

Source- thehindu.com

Q.72) Ans: c

Exp:

Both the statements are correct:

- BharatNet Project** also known as National Optical Fibre Network. It is an ambitious initiative to trigger a broadband revolution in rural areas.
- It aims to connect all the 2,50,000 Gram panchayats in the country and provide 100 Mbps connectivity to all gram panchayats (GPs).
- The entire project is being funded by Universal Service Obligation Fund (USOF), which was set up for improving telecom services in rural and remote areas of the country.
- The objective is to facilitate the delivery of e-governance, e-health, e-education, e-banking, Internet and other services to rural India.
- Bharat net project implemented by Bharat Broadband Network Limited. It is a special purpose vehicle (SPV) set up by the government of India under the Companies Act, 1956.
- Bharat net project comes under the Ministry of Communication and Information Technology.

Source- vikaspedia.com

Extra edge by only IAS
Universal Service Obligation Fund (USOF):

Universal Service Obligation Fund (USOF) is the pool of funds generated by Universal Service Levy (USL).

USL is charged upon all the telecom fund operators on their Adjusted Gross Revenue (AGR).

USL is 05% of AGR of all the telecom operators.

Statutory status to USOF given in 2003 by amendment in **Telegram Act, 1985**.

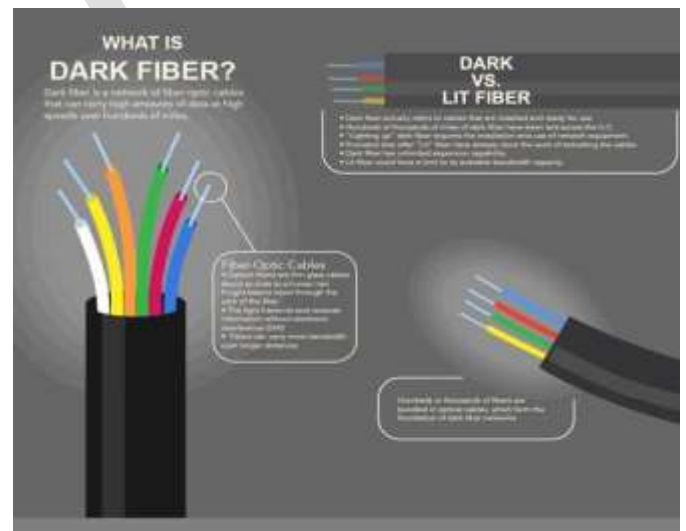
It comes under the Ministry of **Communication and Information Technology**.

Objective- Provide a balance between the provision of Universal Service to all uncovered areas, including the rural areas.

Q.73) Ans: c

Exp:

Dark fibre/Black fibre:





- Dark fibre also known as Unlit Fibre. It is an unused Optical Fibre.
- It has been laid by sector but is not currently being used in fibre optic communication.
- Fibre optic cable transmits information in the form of light pulses, while a dark cable refers to one through which light pulses are not being transmitted.
- Companies lay extra Optical Fibre in order to avoid cost repetition when more bandwidth is needed.
- It will help the telecom service provider to cut down on their capital expenditure.

Q.74) Ans: b

Exp:

- **Statement 1 is incorrect:** Index of Industrial Production data or IIP as it is commonly called is an index that tracks manufacturing activity in different sectors of an economy. The IIP number measures the industrial production for the period under review, usually a month, as against the reference period. IIP is a key economic indicator of the manufacturing sector of the economy. There is a lag of six weeks in the publication of the IIP index data after the reference month ends. IIP index is currently calculated using 2011-2012 as the base year.
- **Statement 2 is correct:** Electricity, crude oil, coal, cement, steel, refinery products, natural gas, and fertilisers are the eight core industries that comprise about 40 percent of the weight of items included in the Index of Industrial Production. Mining, manufacturing, and electricity are the three broad sectors in which IIP constituents fall. The CSO uses secondary data to reach the monthly IIP number. The data is sourced from

various agencies in different ministries or departments of the government. The Department of Industrial Policy and Promotion (DIPP) is the source for the major chunk of data for the calculation.

Source: <https://pib.gov.in/PressReleaseframePage.aspx?PRID=1753865>

Q.75) Ans: b

Exp:

Hydrogen Fuel Cell Vehicles

- Hydrogen Fuel Cell Vehicles (FCVs) are similar to electric vehicles (EVs) in that they use an electric motor instead of an internal combustion engine to power the wheels. However, while EVs run on batteries that must be plugged in to recharge, FCVs generate their electricity onboard.
- In a fuel cell, hydrogen (H₂) gas from the vehicle's fuel tank combines with oxygen (O₂) from the air to generate electricity with only water and heat as byproducts of the process.
- Like electric vehicles, FCVs are zero-emission vehicles – they have no smog-related or greenhouse gas tailpipe emissions. Emissions are created in the process of producing and transporting hydrogen fuel.

Government Missions towards hydrogen fuel

- The Finance Minister in the Union budget for 2020-21 formally announced the National Hydrogen Mission which aims for generation of hydrogen from green power resources.
- The Ministry of New and Renewable Energy (MNRE) has also disclosed that the draft regulations for NHM will be finalised by the end of this month and will thereafter proceed for approval of the Union Cabinet.



Source: <https://pib.gov.in/PressReleaseDetail.aspx?PRID=1752915>

Q.76) Ans: b

Exp:

Stablecoin

- Stablecoins might be the most ironically named innovation of the cryptocurrency era. A stablecoin is a type of cryptocurrency that is typically pegged to an existing government-backed currency.
- The majority of the dozens of stablecoins that currently exist use the dollar as their benchmark asset, but many are also pegged to other fiat currencies issued by governments like the euro and yen.
- As a result, the price of stablecoins fluctuates very little, unlike high-profile cryptocurrencies like bitcoin and ethereum that are prone to sudden ups and downs.
- Stablecoins are useful because they allow people to transact more seamlessly in cryptocurrencies that function as investments, such as Bitcoin. They form a bridge between old-world money and new-world crypto. The first stablecoin, created in 2014, was Tether, which many other stablecoins are modeled after.

Source: <https://www.investopedia.com/terms/s/stablecoin.asp>

Q.77) Ans: c

Exp:

The Global Innovation Index

- The Global Innovation Index is published by World Intellectual Property Organization (WIPO) in

partnership with the Portulans Institute and with the support of corporate network partners. The index ranks 132 world economies according to their innovation capabilities.

- The index ranks world economies according to their innovation capabilities and consists of roughly 80 indicators grouped into innovation inputs and outputs. Innovation inputs are Institutions; Human capital and research; Infrastructure; Market sophistication; Business sophistication and Innovation outputs are Knowledge and technology outputs; Creative outputs.
- Switzerland, Sweden, U.S., and U.K. continue to lead the innovation ranking, and have all ranked in the top 5 in the past three years. The Republic of Korea joins the top 5 of the GII for the first time in 2021.
- India has been ranked 46th in the Report. It was ranked at 48th in 2020. India has been on a rising trajectory over the past several years in the Index.

Source: https://www.wipo.int/global_innovation_index/en/

Q.78) Ans: c

Exp:

- **Statement 1 is correct:** The National Single Window System portal offers a single dashboard, allowing businesses to apply for clearances, track progress and respond to queries. It would become a “one stop shop” for state and Central government compliances and bring transparency, accountability and responsiveness in the ecosystem
- **Statement 2 is correct:** Department for Promotion of Industry and Internal Trade (DPIIT) along with Invest India initiated the process of developing the portal as a National Single Window System (NSWS).



It will also offer a Know Your Approvals service to inform businesses of the details of all the approvals they need to obtain as well as a common registration form, document repository and e-communication module. It will provide strength to other schemes e.g. Make in India, Startup India, PLI scheme etc.

- India has recorded the highest loss in terrestrial water storage if the loss of water storage in Antarctica and Greenland is excluded.
- India is the 'topmost hotspot of TWS loss'. The northern part of India has experienced the maximum loss within the country.

Source: <https://pib.gov.in/PressReleasePage.aspx?PRID=1756966>

Q.79) Ans: d

Exp:

The World Meteorological Organization (WMO) released a report titled the State of Climate Services report 2021. It focuses on Terrestrial Water Storage.

Terrestrial Water Storage (TWS):

- TWS is the sum of all water on the land surface and in the subsurface, i.e. surface water, soil moisture, snow and ice and groundwater.
- Water resources across the world are under tremendous pressure due to human and naturally-induced stressors.
- These include population growth, urbanisation and decreasing availability of freshwater.
- Extreme weather events too have been responsible for the pressure on water resources realised across sectors and regions

Global Scenario:

- TWS dropped at a rate of 1 cm per year in 20 years (2002-2021).
- The biggest losses have occurred in Antarctica and Greenland. But many highly populated, lower latitude locations have also experienced TWS losses.

Indian Scenario:

- The TWS has been lost at a rate of at least 3 cm per year. In some regions, the loss has been over 4 cm per year too.

Extra Edge by Only IAS
World Meteorological Organization(WMO)

WMO is an intergovernmental organization with a membership of 192 Member States and Territories. It originated from the International Meteorological Organization (IMO), which was established after the 1873 Vienna International Meteorological Congress. Established by the ratification of the WMO Convention on 23 March 1950, WMO became the specialized agency of the United Nations for meteorology (weather and climate), operational hydrology and related geophysical sciences. It's Headquartered at Geneva, Switzerland.

Source: <https://reliefweb.int/report/world/2020-state-climate-services-move-early-warnings-early-action>

Q.80) Ans: d

Exp:

Jay Prakash Narayan

- He is also known as "Lok Nayak". People also affectionately called him JP. He was a freedom fighter and one of the pioneers of the socialist movement in India. He was a man of great intellect and ethical values and standards.
- He fought against the repressive practices of Indira Gandhi's government Bhoodan in the 1970s. Furthermore, he also created support for the movement. However, his life had a transition from



socialism to Sarvodaya. This also led to the initiation of the Total Revolution.

- Total Revolution is a revolution against the Indira Gandhi Regime, as she was found guilty of violating electoral laws by the Allahabad High Court. JP advocated a program of social transformation which he termed 'Sampoorna (total revolution)' in 1974 against corruption in public life. This resulted in the defeat of the Indira Gandhi government in 1977 and the installation of the first-ever non-Congress government at the Centre.
- Jayaprakash Narayan was posthumously conferred with India's highest civilian award, the Bharat Ratna in 1999.

Source: <https://www.britannica.com/biography/Jayaprakash-Narayan>

Q.81) Ans: a

Exp:

- The International Energy Agency is a Paris-based autonomous intergovernmental organisation established in the framework of the Organisation for Economic Co-operation and Development (OECD) in 1974 in the wake of the 1973 oil crisis.
- The IEA was initially dedicated to responding to physical disruptions in the supply of oil, as well as serving as an information source on statistics about the international oil market and other energy sectors.
- It is best known for the publication of its annual World Energy Outlook.
- It has 30 members at present. The International Energy Agency family also includes eight association countries. A candidate country must be a country of the OECD (1961). But all OECD members are not IEA members.

- India became an Associate member of IEA in March 2017 but it was in engagement with IEA long before its association with the organization.

Source:

<https://www.bloomberg.com/news/articles/2021-10-14/global-gas-crisis-is-spilling-over-into-oil-markets-iea-says>

Q.82) Ans: d

Exp:

- Plastic packaging, as per the rules divided into three categories.
- **Category 1:** "Rigid" plastic
- **Category 2:** "Flexible plastic packaging of a single layer or multilayer (more than one layer with different types of plastic), plastic sheets and covers made of plastic sheet, carry bags or pouches.
- **Category 3:** It is called multilayered plastic packaging, which has at least one layer of plastic and at least one layer of material other than plastic.
- Producers of plastic would be obliged to declare to the government, via a centralised website, how much plastic they produce annually.
- Companies would have to collect at least 35% of the target in 2021-22, 70 %by 2022-23 and 100
- In 2024, a minimum of 50% of their rigid plastic (category 1) would have to as would 30% of their category 2 and 3 plastic.
- Every year would see progressively higher targets and after 2026-27, 80% of their category 1 and 60% of the other two categories would need to be recycled.
- If entities cannot fulfil their obligations, they would on a "case basis by case " be permitted to buy certificates making up for their shortfall from



organisations that have used recycled content in excess of their obligation.

- The CPCB would develop a “mechanism ” for such a portal .
- Non-compliance would not invite a traditional fine. Instead an “environmental compensation” would be levied, though the rules do not specify how much this compensation would be.

Source: <https://www.thehindu.com/sci-tech/energy-and-environment/centre-announces-plastic-waste-recycling-targets/article36973023.ece>

Q.83) Ans: c

Exp:

Dara Shikoh

- Dara Shikoh (1615-59) was the eldest son and a chosen successor of the Mughal emperor Shah Jahan. He was designated with the title **Padshahzada-i-Buzurg Martaba** (Prince of High Rank). However, he was killed after losing the war of succession against his brother Aurangzeb.
- He is described as a “liberal Muslim” who tried to find commonalities between Hindu and Islamic traditions. His most important works, **Majma-ul-Bahrain** (Mingling of Two Oceans) and **Sirr-i-Akbar** (Great Mystery), are devoted to the cause of establishing connections between Hinduism and Islam, Promotion of Indian Culture.
- He acquired proficiency in Sanskrit and Persian, which enabled him to play a key role in popularising Indian culture and Hindu religious thought.
- He translated the Upanishads and other important sources of Hindu religion and spirituality from Sanskrit to Persian. Through these translations, he was responsible for taking the Hindu culture and spiritual traditions to Europe and the West.

Source: <https://indianexpress.com/article/opinion/colu>

mns/dara-shikoh-never-became-emperor-but-he-was-a-true-child-of-india-7479100/

Q.84) Ans: c

Exp:

- **Statement 1 is correct:** The campaign against lead petrol was led and supported by the UNEP and its Partnership for Clean Fuels and Vehicles (PCFV) in accomplishing the global objective. The use of leaded petrol has been eradicated from the globe, a milestone that will prevent more than 1.2 million premature deaths and save world economies over \$2.4 trillion annually. Algeria is the last country to use the fuel and exhausted its supplies last month. India banned leaded petrol in March 2000.

- **Statement 2 is correct:** The use of lead as a petrol additive has been a catastrophe for public health. Leaded petrol has caused more exposure to lead than any other source worldwide.

Harmful effects of leaded petrol

- o Lead exposure can have serious consequences for the health of children.
- o At high levels of exposure, lead attacks the brain and central nervous system to cause coma and even death.
- o Children who survive severe lead poisoning may be left with mental retardation and behavioral disorders.
- o Lead also causes long-term harm in adults, including increased risk of high blood pressure and kidney damage.

Source: <https://www.downtoearth.org.in/blog/pollution/world-freed-from-toxic-leaded-petrol-a-global-win-78722>

Q.85) Ans: b

Exp:

Universal Postal Union



- Established in 1874, the Universal Postal Union (UPU), with its headquarters in the Swiss capital Bern, is the second oldest international organization worldwide. With its 192 member countries, the UPU is the primary forum for cooperation between postal sector players.
- It helps to ensure a truly universal network of up-to-date products and services. The organization fulfills an advisory, mediating and liaison role, and provides technical assistance where needed.
- It sets the rules for international mail exchanges and makes recommendations to stimulate growth in mail, parcel and financial services volumes and improve quality of service for customers.
- Any non-member country of the United Nations may become a UPU member provided that its request is approved by at least two-thirds of the member countries of the UPU.
- Recently India has won elections for membership of the Council of Administration (CA) and the Postal Operations Council (POC) of Universal Postal Union.

Source: <https://www.upu.int/en/Universal-Postal-Union>

Q.86) Ans: c

Exp:

West Nile virus

- West Nile virus (WNV) is a member of the flavivirus genus. West Nile virus (WNV) is the leading cause of mosquito-borne disease in the continental United States. It is most commonly spread to people by the bite of an infected mosquito.
- Cases of WNV occur during mosquito season, which starts in the summer and continues through fall. There are no vaccines to prevent or medications to treat WNV in people. Fortunately, most people infected with WNV do not feel sick.
- Human infection is most often the result of bites from infected mosquitoes. Mosquitoes become infected when they feed on infected birds, which circulate the virus in their blood for a few days. No human-to-human transmission of WNV through casual contact has been documented.
- About 1 in 5 people who are infected develop a fever and other symptoms. About 1 out of 150 infected people develop a serious, sometimes fatal, illness.

Source: <https://www.cdc.gov/westnile/index.html>

Q.87) Ans: b

Exp:

- **Statement 1 is incorrect:** The Programme for International Student Assessment (PISA) is a triennial international survey which aims to evaluate education systems worldwide by testing the skills and knowledge of 15-year-old students who are nearing the end of their compulsory education. PISA assesses how well they can apply what they learn in school to real-life situations. Over 90 countries have participated in the assessment so far which began in 2000. Every three years students are tested in the key subjects: reading, mathematics and science.
- **Statement 2 is correct:** It is coordinated by the Organization for Economic Cooperation and Development (OECD). India has participated in the PISA test only once before, in 2009. In this India ranked 72nd out of 73 countries, outranking only Kyrgyzstan. Since then, India has stayed away from the test until now, for students from Chandigarh



will be sitting for the test in 2022. Originally the main PISA survey was scheduled for 2021, but was postponed by one year owing to the pandemic.

Source

: <http://pisa.seshagun.gov.in/?AspxAutoDetectCookieSupport=1>

Q.88) Ans: a

Exp:

- Pair 1 is correct: Gharial: It is also known as the gavial or the fish-eating crocodile, It is the longest among all living crocodilians.
- **Conservation:**
 - o Critically Endangered (IUCN Red List).
 - o Schedule I: Indian Wildlife Protection Act, 1972.
 - o Appendix I: CITES.
- **Pair 2 is incorrect:** Mugger Crocodile: Mugger Crocodile is also called marsh crocodile or broad-snouted crocodile. It is an egg-laying and hole-nesting species. It is native to freshwater habitats from southern Iran and Pakistan to the Indian subcontinent and Sri Lanka.
- **Conservation:**
 - o IUCN Status: Vulnerable.
 - o Indian Wildlife Protection Act, 1972: Schedule 1.
 - o CITES: Appendix I.
- Pair 3 is incorrect: Saltwater Crocodile: It is considered as the Earth's largest living crocodile species native to saltwater habitats and brackish wetlands. It is found throughout the east coast of India, Southeast Asia and northern Australia.

Bhitarkanika National Park in Odisha houses 70% of India's estuarine crocodiles or saltwater crocodiles.

• Conservation:

- o IUCN Status: Least Concern
- o Indian Wildlife Protection Act, 1972: Schedule Threats are Illegal hunting, habitat loss and antipathy toward the species because of its reputation as a man-eater.

Source: <https://www.downtoearth.org.in/news/wildlife-biodiversity/odisha-s-kendrapara-now-india-s-only-district-to-have-all-3-species-of-crocodilians-78740>

Q.89) Ans: a

Exp:

- Recently, the India-led United Nations Security Council (UNSC) adopted a Resolution 2593 on Taliban. The resolution called on the Taliban to keep their commitments on preventing terror groups in Afghanistan.
- It urged them to assist the safe evacuations of all Afghan nationals wishing to lead the country. It mentions individuals designated by Resolution 1267 also, which includes the Lashkar E Taiba and Jaish-e-Muhammad.

Source: <https://www.hindustantimes.com/india-news/unsc-resolution-2593-should-shape-world-s-response-to-afghanistan-says-india-101631630677971.html>

Q.90) Ans: d

Exp:

- **Statement 1 is incorrect:** The Deputy Speaker is elected by the Lok Sabha from amongst its members. He/She remains in the office during the



life of Lok Sabha. He/She assumes the power of the speaker when the office is vacant or the speaker is absent. When he/she is appointed as a member of the parliamentary committee, he/she automatically becomes its chairman.

- **Statement 2 is incorrect:** Article 93 of the constitution provides for the election of both the Speaker and the Deputy Speaker. The date of election of the Deputy Speaker is fixed by the Speaker. He/She can vote at the first instance. But when acting as the Speaker, he/she has the privilege of the Casting vote in case of tie. The Deputy Speaker gives the resignation letter to the Speaker and can be removed in the same manner as the Speaker.

Source: <https://www.thehindu.com/news/national/long-wait-for-a-deputy-speaker-for-lok-sabha/article36252613.ece>

Q.91) Ans: a

Exp:

- **Statement 1 is correct:** Pulicat Lake is the second-largest brackish water lake in the country, next only to Chilika lake. Also called Pazhaverkadu. The region in which Pulicat lake is situated receives rainfall from both South-West as well as North-East monsoon winds.
- **Statement 2 is incorrect:** It lies at the border of the states of Andhra Pradesh and Tamil Nadu on the Barricade Island of Sriharikota. Two rivers that feed the lagoon are the Arani River at the southern tip and the Kalangi River from the northwest. Nelapattu Bird Sanctuary is a famous bird sanctuary located near this lake. The lake is also home to black-headed ibis, Asian openbill, black-crowned night heron, and little cormorant. The project "Mangrove Restoration in Asia" is

implemented at Pulicat Lake. Flamingo Festival is held every year to promote tourism in Pulicat and Nelapattu.

•

Source : https://en.wikipedia.org/wiki/Pulicat_Lake

Q.92) Ans: b

Exp:

- Established in the Hindu Kush in 1893, Durand Line runs through the tribal lands between Afghanistan and British India. In modern times it has marked the border between Afghanistan and Pakistan.
- In the 19th century Afghanistan was used as a buffer by the British against the fear of Russian expansionism to its east. Hence the British civil servant Sir Henry Mortimer Durand signed an agreement with Afghan ruler Amir Abdur Rahman on November 12, 1893 demarcating the Durand line.
- The line stretches from the border with China to Afghanistan's border with Iran. This line demarcated the limits of Afghanistan's and British India's "spheres of influence" on the Afghan "frontier" with India.
- The line puts the strategic Khyber Pass on the British side. It cuts through Pashtun tribal areas, leaving villages, families, and land divided between the two spheres of influence.

Source: https://en.wikipedia.org/wiki/Durand_Line

Q.93) Ans: a

Exp:



- **Statement 1 is correct:** EEF was established by the decree of the President of the Russian Federation in the year 2015. Its aim is to support the economic development of Russia's Far East and to expand international cooperation in the Asia-Pacific region. It takes place each year in Vladivostok, a city in Russia.
- **Statement 2 is incorrect:** It serves as a platform for the discussion of key issues in the world economy, regional integration, and the development of new industrial and technological sectors, as well as of the global challenges facing Russia and other nations. The Forum's business programme includes a number of business dialogues with leading partner countries in the Asia-Pacific region, and with ASEAN, a key integration organization of dynamically developing nations in Southeast Asia. Over the years, it has emerged as an international platform for discussing the strategy for developing political, economic and cultural ties between Russia and Asia Pacific.

Source: <https://forumvostok.ru/en/>

Q.94) Ans: b

Exp:

Inspiration4

- Inspiration4 is the world's first all-civilian mission to orbit. The flight will be privately operated by SpaceX (US Private Space Company). Inspiration4 will blast off from NASA's Kennedy Space Center in Florida. The mission involves circling the Earth for three days, and then splashing down into the Atlantic Ocean.
- Inspiration4 will orbit the Earth at 575km, higher than the International Space Station (408km) and the Hubble space telescope (547km). This will be

the farthest distance travelled by a crewed mission since 2009, when astronauts last went to repair the Hubble. The mission will raise awareness and funds for St. Jude Children's Research Hospital in the US.

- The mission involves circling the Earth for three days, and then splashing down into the Atlantic Ocean. The journey will present an opportunity for collecting large amounts of health data that will aid in planning future crewed space missions. Inspiration4 is seen as part of an effort to open up space travel to non-professionals.

Source: <https://indianexpress.com/article/explained/inspiration4-spacex-first-all-civilian-space-mission-7490823/>

Q.95) Ans: a
Exp:

- **Statement 1 is correct:** Karbi Anglong Agreement is a tripartite agreement signed by the Centre, five insurgent groups active in Karbi Anglong, and the Assam government, marks the culmination of an extended process of negotiation to end insurgency in the region. Karbi Anglong is the largest district in Assam. The district is administered by Karbi Anglong Autonomous Council according to the Sixth Schedule of the Constitution of India. The district comprises various tribal and ethnic groups including the Kukis, Dimasas, Garos, Rengma Nagas, Tiwas and Karbis.
- **Statement 2 is incorrect:** It provides greater autonomy to the Karbi Anglong Autonomous Council; the identity, language, culture of Karbi people will be protected; and more focused development will be carried out in the Council area. The government will also consider notifying Karbi language as the official language of the Council.



Over a thousand armed insurgents have surrendered their arms under the peace deal. The Government of Assam will set up a Karbi Welfare Council to focus on the development of the Karbi people living outside the Karbi Anglong Autonomous Council area. The Consolidated Fund of the State will be amended to meet the resources of the Karbi Anglong Autonomous Council.

Source :
<https://indianexpress.com/article/opinion/editorials/as-sam-government-karbi-anglong-insurgent-groups-7493013/>

Q.96) Ans: a

Exp:

- The Astrophysical Journal identifies a new class of exoplanets: Hycean worlds that could support life different from that on Earth. The word Hycean comes from the words hydrogen and ocean. Planet-wide oceans and hydrogen-rich atmospheres might cover these worlds. They are also uniquely alien, up to 2.6 times the diameter of Earth, with temperatures up to 200 degrees celsius and thick hydrogen atmospheres. This places them somewhere between Earth and giant planets like Neptune or Uranus. These planets are variedly classed as super-Earths or mini-Neptunes depending on inferences about their bulk compositions based on their densities.

Exoplanet:

- An exoplanet is any planet beyond our solar system. More than 4,400 planets have been discovered outside our solar system.
- Most orbit other stars, but free-floating exoplanets, called rogue planets, orbit the galactic centre and are untethered to any star.

Source: <https://www.airspacemag.com/daily-planet/hycean-worlds-new-type-ocean-planet-180978567/>

Q.97) Ans: b

Exp:

- **Statement 1 is incorrect:** The 13th BRICS Summit will be held under India's Chairship. It will be the third time that India will be hosting the BRICS Summit after 2012 and 2016. The theme for India's Chairship is 'BRICS @ 15: Intra-BRICS Cooperation for Continuity, Consolidation and Consensus'. As BRICS celebrates its 15th anniversary this year. It is a matter of great pride that BRICS has been a beacon of multilateralism based on equality, mutual respect and trust.
- **Statement 2 is correct:** BRICS is an important grouping bringing together the major emerging economies from the world, comprising 41% of the world population, having 24% of the world GDP and over 16% share in the world trade. BRICS countries have been the main engines of global economic growth over the years. Over a period of time, BRICS countries have come together to deliberate on important issues under the three pillars of political and security, economic and financial and cultural and people to people exchanges.

Source : <https://brics2021.gov.in/>

Q.98) Ans: b

Exp:

- **Statement 1 is incorrect:** The dugong (Dugong dugon), also called the sea cow, is a herbivorous mammal. They are found in over 30 countries and in India are seen in the Gulf of Mannar, Gulf of Kutch, Palk Bay, and the Andaman and Nicobar



Islands. Dugongs are listed as Vulnerable on the IUCN Red List of Threatened Species.

- Statement 2 is correct: The loss of seagrass habitats, water pollution and degradation of the coastal ecosystem due to developmental activities have made life tough for these slow-moving animals. Dugongs are also victims of accidental entanglement in fishing nets and collision with boats, trawlers. Internationally, dugongs are listed in Appendix I of the Convention on International Trade in Endangered Species of Wild Fauna and Flora (CITES), which prohibits the trade of the species and its parts. India's first dugong conservation reserve is set up by the Tamil Nadu State government in the Palk Bay.

Source

: <https://indianexpress.com/article/technology/science/indias-first-dugong-conservation-reserve-7494506>

Q.99) Ans: c

Exp:

- **Statement 1 is correct:** National Institute Ranking Framework or NIRF is the first-ever effort by the government to rank higher education institutions (HEIs) in the country. Before NERFs launch in 2016, HEIs were usually ranked by private entities, especially news magazines. While participation in the NIRF was voluntary in the initial years, it was made compulsory for all government-run educational institutions in 2018.
- **Statement 2 is correct:** IIT-Madras, IISc-Bangalore, and IIT-Bombay have emerged as the country's top three higher education institutions. Under the best college category, Miranda House has again topped the list this year. In the university category, IISc Bengaluru was the best-ranked varsity. NIRF is in its sixth year, but it continues to only rank Indian HEIs.

Source

: <https://indianexpress.com/article/explained/what-is-nirf-and-why-did-the-government-decide-to-rank-universities-7499133/>

Q.100) Ans: b

Exp:

Pradhan Mantri Gramin Digital Saksharta Abhiyan

- PMGDISHA is expected to be one of the largest digital literacy programmes in the world. Scheme is applicable only for rural areas of the country.
- It envisages to make 6 crore persons in rural areas, across States/UTs, digitally literate. This will reach to around 40% of rural households by covering one member from every household where there is no digitally literate person.

Entry criteria

- Beneficiaries should be Digitally Illiterate, who are only from rural areas of the country.
- All such households where none of the family members are digitally literate will be considered as eligible households under the Scheme.
- Only one person per eligible household would be considered for training
- Age Group - 14 to 60 years

Priority would be given to:

- Non-smartphone users, Antyodaya households, college drop-outs, Participants of the adult literacy mission.
- Digitally illiterate school students from class 9th to 12th, provided facility of Computer/ICT Training is not available in their schools.
- Preference would be given to SC, ST, BPL, women, differently-abled persons and minorities.



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- Training will be given in the nearest Training Centre or CSCs for 20 hours which has to be completed in a minimum of 10 days and maximum of 30 days.

Source: <https://vikaspedia.in/e-governance/digital-india/national-digital-literacy-mission>

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